A COMPLETE PROJECT BOOK

Beyond Accuracy

A Leakage-Audited, Explainable and Uncertainty-Aware Stacking Ensemble for Postpartum Anxiety Screening from Questionnaire Data

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Interpretable machine learning · leakage-aware evaluation · health survey data

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Beyond Accuracy: A Leakage-Audited, Explainable and Uncertainty-Aware Stacking Ensemble for Postpartum Anxiety Screening from Questionnaire Data

Motivation, methods, results, limitations and future directions of a seven-model study on 1,503 questionnaire responses

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SOURCE OF NUMBERS	Every table and figure in this book is produced by the study notebook. The book is generated from the notebook's result files, so the text and the results cannot drift apart.
STATUS	Research result. Not a medical device and not for individual diagnosis.

BEFORE CITING THIS WORK

Accuracies of earlier papers were taken from abstracts and search summaries, not from the full texts. Entries marked confirm in the reference list have details that could not be verified. Open each paper and check it before you cite it.

Abstract

Background. Machine-learning studies on the public *PostPartum Depression* questionnaire dataset report accuracies between 95.68% and 99.6%. They rarely explain individual predictions, quantify uncertainty, test whether one model is really better than another, or examine the fact that most rows of the dataset repeat an earlier answer pattern.

Objective. To build and evaluate an explainable stacking ensemble that predicts a self-reported *Feeling anxious* answer from eight other questionnaire answers and an age band, and to audit how far the reported accuracy reflects prediction on new answer patterns.

Methods. All 1,503 responses were split into a stratified 1,202-person training set and a 301-person test set. Seven models competed: logistic regression, random forest, XGBoost, LightGBM, CatBoost, a multilayer perceptron, and a proposed Hybrid Stack in which CatBoost, XGBoost, random forest and the perceptron feed a logistic-regression meta-learner through out-of-fold predictions. Models were compared with ten-fold cross-validation, a single final test, the Friedman test, Wilcoxon signed-rank tests with Holm correction, McNemar tests and a bootstrap confidence interval. The Hybrid Stack was explained with SHAP and LIME, which were compared on agreement, deletion-based faithfulness, stability and speed. Calibration, conformal prediction, a counterfactual example and an age-group check were added. Finally, grouped cross-validation kept every answer pattern on one side of each split.

Results. The Hybrid Stack reached 98.67% test accuracy (95% CI 97.34–99.67%), F1 98.97% and ROC-AUC 1.000. CatBoost reached 100.00% on the same split and 98.25% ± 1.44 in cross-validation. The Hybrid Stack was significantly better than logistic regression (Holm-corrected $p = 0.012$) and not significantly different from the other five models. SHAP and LIME both ranked *Feeling of guilt* first (37.3% and 38.1% of total importance); they agreed globally (Spearman 0.917) but less per person (0.534), and SHAP was more faithful. Conformal prediction at a 95% target covered 97.01% of test people. However, 93.4% of test people had an identical answer pattern in the training set; with grouped cross-validation, accuracy fell to 80.2–86.0% for the six non-linear models.

Conclusion. Under the random-split protocol used in earlier work, the proposed ensemble matches the published range, with explanations, uncertainty estimates and statistical tests that earlier studies lack. The duplicate audit shows that the 98–100% figures on this dataset mostly measure recognition of answer patterns already seen, and that about 85% is the realistic estimate for a new pattern. Future studies on this dataset should report duplicate-aware validation next to the standard one.

KEYWORDS postpartum anxiety · perinatal mental health · stacking ensemble · explainable AI · SHAP · LIME · conformal prediction · data leakage · duplicate-aware validation · questionnaire data

Key findings at a glance

98.67%
HYBRID STACK, TEST SET
95% CI 97.34–99.67%; 4 errors in 301 people

98.00%
HYBRID STACK, 10-FOLD CV
± 1.26 points across folds

84.7%
ON UNSEEN ANSWER PATTERNS
Grouped 5-fold CV; the realistic estimate

93.4%
TEST PEOPLE WITH A TWIN
Identical answer pattern already in the training set

0.917
SHAP-LIME GLOBAL AGREEMENT
Per person only 0.534

97.0%
CONFORMAL COVERAGE
Against a 95% promise

One model, four ways of measuring it

Protocol	Accuracy
Held-out test set (random split)	98.67%
Ten-fold cross-validation (random folds)	98.00%
Grouped cross-validation (unseen patterns)	**84.70%**
Always answering "anxious" (floor)	65.20%

The seven models side by side

Model	10-fold CV	Test	Grouped CV	Test errors	Brier
CatBoost	98.25%	100.00%	85.16%	0	0.002
Random Forest	97.92%	99.34%	86.03%	2	0.014
MLP Neural Net	97.59%	98.67%	80.17%	4	0.015
Hybrid Stack (proposed)	**98.00%**	**98.67%**	**84.70%**	**4**	**0.006**
LightGBM	98.08%	97.67%	83.70%	7	0.022
XGBoost	97.83%	97.01%	83.83%	9	0.020
Logistic Regression	82.70%	81.73%	79.84%	55	0.122

IN ONE SENTENCE

The proposed ensemble is as accurate as the best published models under their protocol, explains itself and knows when it is unsure, and the audit behind it shows that on this dataset about 13 points of every reported accuracy come from repeated answer patterns.

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CHAPTER 1

Introduction

Why another model for a dataset that already reports 99% accuracy, and what this project sets out to show instead.

1.1 Why postpartum anxiety matters

The weeks and months after childbirth are a period of high risk for a mother's mental health. Postpartum depression is the best-known condition, but anxiety in the same period is also common, frequently occurs together with depression, and is easier to miss because sleeplessness, worry and irritability are often dismissed as a normal part of new parenthood. Untreated, perinatal anxiety and depression affect the mother, her relationship with her partner and the early bond with the baby.

Screening is the first line of defence. In practice it is done with short questionnaires: a mother answers a handful of questions, and the answers decide whether a clinician looks more closely. A questionnaire is cheap, quick and can be given anywhere, which is exactly why researchers have asked whether a machine-learning model could read the answers and flag women at risk automatically. A growing body of work, reviewed in Chapter 3, has trained models on hospital records, cohort studies and questionnaire data [6–13].

This project works with one such questionnaire dataset: 1,503 responses published on Kaggle under the name *PostPartum Depression* [30]. It contains an age band and nine items about mood, sleep, appetite, guilt, bonding with the baby and suicide attempt. One item, *Feeling anxious*, is the target that the dataset's own description names, and it is the answer this study predicts.

1.2 The problem with accuracy alone

Several papers have used the same 1,503 rows and report accuracies from 95.68% to 99.6% [1–3, 5]. On the surface the problem looks solved. Four things are missing, however, before such a number can support a screening tool.

- **An explanation.** A clinician cannot act on "the model says anxious" without knowing which answers produced that result. Most earlier studies on this dataset give no per-person explanation, and the one that shows several explanation methods side by side does not measure them against each other [4].
- **An honest statement of uncertainty.** Every prediction in those papers is delivered with full confidence. A screening tool should be able to say "I am not sure" and hand the case to a human.
- **Evidence that the winning model really wins.** A single accuracy on a single test split of about 300 people cannot separate 98.7% from 99.6%. Without a confidence interval and a statistical test, a ranking of models is an anecdote.

- **A check on the data itself.** Of the 1,503 responses in this dataset, 1,177 repeat an answer pattern that already occurs earlier in the file. A random split therefore puts copies of the same pattern in both the training and the test set, and a model can score highly by recognising what it has already seen.

> **THE CENTRAL IDEA OF THIS BOOK**
>
> A high accuracy is where the evaluation of a screening model starts. This project builds a competitive model and then asks four further questions of it: can it explain itself, does it know when it is unsure, is it really better than its rivals, and does its accuracy survive when it is tested on answer patterns it has never seen?

1.3 Research questions

The study is organised around five research questions.

Table 1.1. Research questions and the chapters that answer them.

	Research question	Answered in
RQ1	Which of seven models, from a linear baseline to a stacked hybrid, predicts the *Feeling anxious* answer best, and are the differences statistically significant?	Chapter 6
RQ2	Which questionnaire answers drive the hybrid model's predictions, and do SHAP and LIME give the same explanation?	Chapter 7
RQ3	When SHAP and LIME disagree, which is more faithful to the model, more stable and cheaper to compute?	Chapter 7
RQ4	Can the model give calibrated probabilities, valid uncertainty sets, actionable counterfactuals and comparable accuracy across age groups?	Chapter 8
RQ5	How much of the measured accuracy remains when no answer pattern is shared between training and test data?	Chapter 9

1.4 Objectives

1. Build seven classifiers under one identical preprocessing and validation protocol so that their results are directly comparable.
2. Propose and evaluate a heterogeneous stacking ensemble (the Hybrid Stack) that combines two boosting models, a bagging model and a neural network.
3. Support every comparison with a confidence interval and an appropriate statistical test.
4. Explain the hybrid's predictions with two independent methods and measure those methods against each other.
5. Add uncertainty quantification, a counterfactual explanation and a subgroup check.
6. Audit the dataset for repeated answer patterns and report a duplicate-aware accuracy next to the standard one.
7. Keep the whole study reproducible: fixed seeds, pinned library versions, one notebook of 57 small tasks.

1.5 Contributions

The project makes five contributions.

- **A competitive, fully documented stacking ensemble.** The Hybrid Stack reaches 98.67% test accuracy with a ROC-AUC of 1.000 and one of the best-calibrated probability outputs among the seven models.
- **A statistically honest model comparison.** The tests show that six of the seven models are indistinguishable on this data, a finding that single-number comparisons in earlier work could not reveal.
- **A quantitative SHAP-versus-LIME comparison.** The two methods are measured on agreement, faithfulness, stability and speed, instead of only being displayed.
- **Trustworthy-AI additions that are new for this dataset.** Conformal prediction sets, a counterfactual example and an age-group accuracy check.
- **A duplicate audit.** To our knowledge from the abstracts reviewed, no earlier study on this dataset analyses its repeated answer patterns. This book shows that they explain most of the gap between about 85% and 98–100%.

1.6 How this book is organised

Chapter 2 introduces the concepts a reader needs. Chapter 3 reviews earlier work and states seven research gaps. Chapter 4 describes the dataset. Chapter 5 gives the full methodology with diagrams. Chapters 6 to 10 present the results: the model competition, the explanations, the trustworthy-AI checks, the duplicate audit and the comparison with published work. Chapter 11 discusses what the results mean and what was solved, Chapter 12 lists the limitations, Chapter 13 sets out future work and Chapter 14 concludes. Appendices give the reproduction steps, a map of the notebook, complete result tables and a glossary.

CHAPTER 2

Background and Core Concepts

The ideas the study relies on, explained without assuming prior knowledge of any of them.

2.1 Screening is not diagnosis

A **diagnosis** is made by a clinician after an interview and, where needed, further assessment. A **screening** result only sorts people into "look closer" and "probably fine". This difference shapes how a screening model should be judged. Missing a woman who needs help (a false negative) is usually worse than asking a healthy woman a few more questions (a false positive), so recall matters as much as accuracy. It also means the label in a screening dataset is not a diagnosis. In this project the label is one self-reported questionnaire item, which limits what any model trained on it can claim.

2.2 Tree ensembles, boosting and neural networks

The seven models in this study belong to four families.

- **Linear models.** Logistic regression adds up a weight for each answer and converts the sum into a probability. It is transparent and fast, but it cannot represent interactions, such as an answer that matters only in combination with another.
- **Bagging.** A random forest trains hundreds of decision trees on random resamples of the data and averages their votes. Averaging many independent trees gives a stable model that rarely overfits badly.
- **Boosting.** XGBoost [19], LightGBM [20] and CatBoost [21] also build many trees, but one after another: each new tree concentrates on the mistakes of the trees before it. The three differ in how they grow trees and handle categories. CatBoost was designed with categorical data in mind, which suits questionnaire answers.
- **Neural networks.** A multilayer perceptron (MLP) passes the inputs through layers of weighted sums and non-linear functions. With two hidden layers it is a small representative of deep learning.

2.3 Stacked generalisation

Different model families make different mistakes. **Stacking**, introduced by Wolpert [18], exploits this. A set of *base models* each produce a probability, and a second model, the *meta-learner*, learns how to combine them. The important detail is how the meta-learner is trained. If it saw base-model predictions for the same people those models were trained on, it would learn to trust models that have

merely memorised. Stacking therefore uses **out-of-fold predictions**: the training set is cut into folds, and each base model predicts only the fold it was not trained on. The meta-learner sees honest predictions and learns realistic weights.

With base models $f_1, \ldots, f_4$ and a logistic-regression meta-learner, the final probability for a person with answers x is

$$P(\text{anxious} \mid x) = \sigma\left(w_0 + w_1 f_1(x) + w_2 f_2(x) + w_3 f_3(x) + w_4 f_4(x) \right),$$

where $\sigma(z) = 1 / (1 + e^{-z})$ is the logistic function and the weights w are learned from the out-of-fold probabilities.

2.4 Explaining a prediction: SHAP and LIME

An ensemble of hundreds of trees and a neural network cannot be read like a formula. Explainable AI (XAI) methods give a post-hoc account of a single prediction: how much did each answer push the probability up or down?

SHAP

SHAP [14] is based on Shapley values from cooperative game theory. It treats the answers as players in a game whose payout is the model's prediction and asks what each player contributed on average over all possible orders in which the players could join. The contribution of answer j is

$$\varphi_j = \sum_{S \subseteq F \setminus \{j\}} \left[|S|!\,(|F| - |S| - 1)! \,/\, |F|! \right] \left(v(S \cup \{j\}) - v(S) \right),$$

where F is the set of all features and $v(S)$ is the model's expected output when only the answers in S are known. SHAP values have a property that makes them easy to check: the baseline prediction plus all contributions equals the model's output for that person exactly. With nine features the exact computation is feasible, so this study uses exact SHAP and no approximation.

LIME

LIME [15] takes a different route. Around the person to be explained it generates many slightly changed versions of the answers, asks the model for a prediction on each, and fits a simple weighted linear model to those predictions. The weights of that local linear model are the explanation. LIME is fast and works with any model, but its output depends on random sampling, and its weights do not add up to the prediction.

Why compare them

The two methods can disagree, and when they do a user has no way of knowing which to trust. This is known as the disagreement problem [16]. Chapter 7 therefore does not only show both explanations; it measures how often they agree, which one finds the answers that really move the model, and which one is more stable.

2.5 Honest uncertainty: calibration and conformal prediction

A probability is **calibrated** if, among all people given a predicted probability of 0.8, about 80% really are positive. The **Brier score**, the mean squared difference between predicted probability and outcome, summarises calibration and accuracy together; lower is better.

Conformal prediction [23, 24] goes one step further. Instead of a single answer it returns a *set* of answers with a guarantee: if the user asks for 95% coverage, the true answer lies in the set at least about 95% of the time, whatever model is underneath. For a two-class problem the set can be one answer (the model commits), both answers (the model is unsure), or empty (no answer is plausible enough at this strictness). The method needs a nonconformity score for each calibration person, here one minus the probability the model gave to that person's true class, and a threshold taken from a quantile of those scores.

2.6 Counterfactual explanations

SHAP and LIME describe why the model said what it said. A **counterfactual explanation** [25] answers a more practical question: what is the smallest change to the answers that would have changed the prediction? For a woman flagged as anxious, a counterfactual such as "if irritability were *sometimes* instead of *yes*, the model would no longer flag her" tells a clinician which symptom the model's decision hinges on.

2.7 Data leakage and repeated records

Data leakage is any route by which information from the test data reaches the model during training. It makes a model look better than it is and is a leading cause of irreproducible results in machine-learning-based science [27]. Common routes are fitting a scaler or encoder on all data before splitting, tuning hyper-parameters on the test set, and **repeated records**: if the same row appears on both sides of a split, the test no longer measures prediction on new cases.

The remedy for repeated records is **grouped validation**. Rows that share the same content are given the same group label, and the split is made so that a whole group falls on one side. The test set then contains only patterns the model has never seen, which is the situation a deployed screening tool faces when a woman gives a new combination of answers.

CHAPTER 3

Related Work and Research Gaps

What earlier studies did with this dataset and with postpartum data in general, and the seven gaps they leave open.

3.1 Studies on the same dataset

Five published studies use the same 1,503-row questionnaire dataset. Table 3.1 summarises them. Details marked confirm could not be verified from the abstract and must be checked in the full text.

Table 3.1. Studies on the Kaggle *PostPartum Depression* dataset. Results as reported in abstracts and search summaries.

Study	Method	Reported result	XAI	What is missing
Lilhore et al., 2024 [1]	Hybrid transfer learning with BiLSTM and CNN, compared with decision tree, random forest, AdaBoost, LightGBM, XGBoost and CatBoost	Accuracy 99.6%, precision 99.7%	Not reported in the abstract confirm	No explanation of predictions, no uncertainty, repeated answer patterns not discussed
Raza et al., 2024 [2]	MDKR: decision tree, KNN and random forest feeding an MLP meta-learner; eleven algorithms compared with k-fold validation	Accuracy 99%	Not reported in the abstract confirm	As above. Some sources say the suicide-attempt item was the target confirm
Nayee et al., 2026 [3]	SVM, XGBoost, random forest, decision tree and RBFNN/KNN with recursive feature elimination	Accuracy 96.99%	No	No XAI, no statistical tests, no duplicate analysis
Shivaprasad et al., 2024 [4]	Custom ensemble stacking of baseline classifiers; target reported as the suicide-attempt item confirm	Not confirmed confirm	SHAP, LIME, ELI5, Anchor	Different target; XAI methods shown side by side but apparently not compared numerically
JISEM study [5] confirm	Random forest, XGBoost, CatBoost	CatBoost 95.68%, XGBoost 95%, RF 94%	Not confirmed	No statistical tests or uncertainty in the summary seen

A sixth item, a 2025–26 preprint on explainable deep learning for postpartum depression (accuracy 96%, ROC-AUC 0.980), may use a different 1,503-record dataset and is therefore not included in the numerical comparison.

The benchmark on this dataset is thus 95.68% to 99.6% accuracy under a random train/test split of all rows. Three observations follow. First, the range is narrow and close to the ceiling, so further gains in accuracy are small and hard to demonstrate. Second, the studies report single numbers without confidence intervals. Third, none of the abstracts mentions that most rows are repeats, which Chapter 9 shows is essential for interpreting any of these numbers.

3.2 The wider postpartum machine-learning literature

Studies on other, usually larger and clinically labelled, data report far more modest performance (Table 3.2). Areas under the ROC curve of 0.72 to 0.84 are typical for electronic-health-record cohorts, and two recent reviews conclude that validation is inconsistent and the risk of bias is high [12, 13]. The contrast with the 96–99.6% reported on the Kaggle questionnaire is itself a signal that the latter is an unusually easy benchmark.

Table 3.2. Selected machine-learning studies on postpartum depression using other data.

Study	Data	Result	Note
Zhang et al., 2020 [6]	508 women, China	AUC 0.78	Small cohort
Shin et al., 2020 [7]	PRAMS survey, 28,755 records	Not extracted	PHQ-2 label
Amit et al., 2021 [8]	266,544 UK primary-care records	AUC 0.72–0.74 (0.84 with EPDS)	Outcome from health records
Saqib et al., 2021 [9]	Scoping review, 14 studies	AUC 0.78–0.93	SVM and random forest most common
Nakamura et al., 2024 [10]	Japanese cohort	AUC 0.83	Single site
Huang et al., 2025 [11]	1,065 women	AUC 0.955 (10-fold CV)	Eight algorithms with SHAP; internal validation only
Alkhateeb et al., 2026 [12]	Scoping review, 65 studies	Not applicable	Classical ML dominant; inconsistent validation
Xia et al., 2025 [13]	Systematic review	Not applicable	High risk of bias, limited generalisability

3.3 Seven research gaps

Reading the two bodies of work together gives seven gaps. Each is a concrete thing that earlier studies on this dataset do not report and that this project adds.

Table 3.3. Research gaps, the evidence for each, and how this project addresses it.

Gap	What is missing	Evidence	What this project does	Chapter
G1	High accuracy from black-box models	Studies [1–3] give no per-person explanation	Explains every prediction of the hybrid with SHAP and LIME	7
G2	XAI methods are shown, not compared	Study [4] displays several XAI outputs	Measures SHAP against LIME on agreement, faithfulness, stability and speed	7
G3	No uncertainty: every prediction is given with full confidence	None of [1–4] reports it	Calibration analysis and conformal prediction sets	6, 8

Gap	What is missing	Evidence	What this project does	Chapter
G4	No actionable explanation	None of [1–4]	Counterfactual "what would have to change" example	8
G5	Superiority claimed without statistical tests	[1–3] report single numbers	Friedman, Wilcoxon–Holm, McNemar and a bootstrap confidence interval	6
G6	Repeated answer patterns are not analysed	Not mentioned in any abstract	Reports standard results and duplicate-aware grouped validation	9
G7	No subgroup check	None of [1–4]	Accuracy in every age group	8

3.4 Where this project stands

On raw accuracy this project is in the same range as the best earlier papers and does not claim to exceed the 99–99.6% ones. Its strength lies in what surrounds the number. Table 3.4 puts the two side by side.

Table 3.4. Earlier studies on the dataset compared with this project.

Aspect	Earlier studies [1–5]	This project
Accuracy, random split	95.68% – 99.6%	Hybrid Stack 98.67% (95% CI 97.34–99.67%); CatBoost 100.00%; 10-fold CV 97.6–98.2% for the six non-linear models
Confidence interval and tests	Not reported in the abstracts	Bootstrap CI, Friedman, Wilcoxon–Holm, McNemar
Explanation of each prediction	Only [4]	SHAP and LIME for 100 test people
XAI methods measured against each other	None found	Agreement, faithfulness, stability, speed
Uncertainty of a prediction	None found	Conformal prediction and calibration
Counterfactual explanation	None found	Yes
Repeated answer patterns analysed	None found	Yes; grouped CV gives 84.7% for the hybrid

CHAPTER 4

The Dataset

Where the data come from, what each question looks like, and the one property of the file that changes how every result must be read.

4.1 Source and structure

The data are the public Kaggle dataset *PostPartum Depression* [30]. According to its listing, the answers were collected with an online form at a medical hospital. The file has 1,503 rows and 11 columns: a timestamp, an age band and nine questionnaire items. Table 4.1 is the data card.

Table 4.1. Data card.

Property	Value
Source	Kaggle, *PostPartum Depression*, uploader `parvezalmuqtadir2348`
Collection	Per the listing: questionnaire given through an online form at a medical hospital
Size	1,503 rows, 11 columns (timestamp, age band, nine items)
Timestamps	From 14 June 2022, 20:02 to 15 June 2022, 22:24: all rows fall within 26.4 hours
Target	*Feeling anxious* (Yes / No), the target named on the listing
Features used	9: age band and the eight remaining items
Missing values	27 cells in three columns, kept as a separate "No answer" category
Repeated rows	1,177 rows repeat an earlier answer pattern; 326 patterns are unique
Licence	Listed as MIT in a search result for the Kaggle page; confirm on the page before redistribution
Sensitivity	Self-reported mental-health items, including a suicide-related question

AN OBSERVATION ABOUT PROVENANCE

The timestamps place all 1,503 responses inside a single window of 26.4 hours. The book does not speculate about why, but together with the repeated answer patterns it means the file should be treated as a benchmark of uncertain origin and not as a documented clinical sample.

4.2 The questions and their answers

All features are categorical with three or four possible answers. Table 4.2 lists every answer, how often it was given and the share of women giving it who also answered *Yes* to *Feeling anxious*.

Table 4.2. Every feature, its answers, their frequency and the anxious rate among women who gave that answer.

Question	Answer	Count	Share	Anxious
Age	40-45	364	24.2%	62.4%
	35-40	349	23.2%	66.2%
	30-35	338	22.5%	66.0%
	45-50	271	18.0%	72.0%
	25-30	181	12.0%	57.5%
Feeling sad or Tearful	Yes	536	35.7%	64.7%
	No	524	34.9%	69.3%
	Sometimes	443	29.5%	60.9%
Irritable towards baby & partner	Yes	555	36.9%	76.9%
	No	499	33.2%	74.3%
	Sometimes	443	29.5%	39.7%
	No answer	6	0.4%	100.0%
Trouble sleeping at night	Two or more days a week	640	42.6%	58.1%
	Yes	445	29.6%	62.9%
	No	418	27.8%	78.5%
Problems concentrating or making decision	No	583	38.8%	48.9%
	Often	473	31.5%	84.4%
	Yes	435	28.9%	65.3%
	No answer	12	0.8%	100.0%
Overeating or loss of appetite	No	841	56.0%	68.8%
	Yes	343	22.8%	48.1%
	Not at all	319	21.2%	74.0%
Feeling of guilt	No	624	41.5%	89.3%
	Maybe	528	35.1%	64.6%
	Yes	342	22.8%	21.3%
	No answer	9	0.6%	100.0%
Problems of bonding with baby	No	557	37.1%	48.7%
	Sometimes	542	36.1%	75.5%

Question	Answer	Count	Share	Anxious
	Yes	404	26.9%	74.3%
Suicide attempt	No	709	47.2%	73.2%
	Yes	459	30.5%	72.1%
	Not interested to say	335	22.3%	38.8%

One pattern in the table deserves attention. For *Feeling of guilt*, 89.3% of the women who answered *No* report feeling anxious, against 64.6% for *Maybe* and only 21.3% for *Yes* (Figure 4.1). The relationship is strong, which is why this question dominates every importance ranking later in the book, but its direction is the opposite of what clinical experience would suggest. It must be read as a property of this dataset, not as evidence about guilt and anxiety in general.

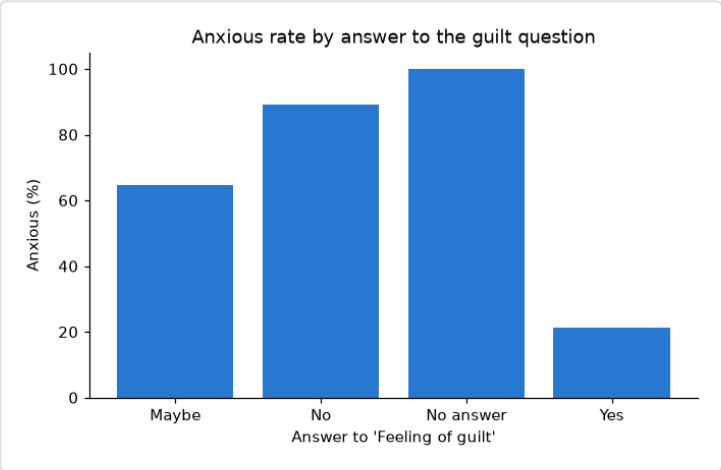


Figure 4.1. Share of women answering *Yes* to *Feeling anxious*, by their answer to the guilt question.

4.3 Missing answers

Only 27 of the cells are empty: 6 in *Irritable towards baby & partner*, 12 in *Problems concentrating or making decision* and 9 in *Feeling of guilt*. They are not imputed. A skipped question can itself carry information, so every empty cell becomes its own answer category, "No answer". No row is dropped.

4.4 The target and its balance

The target is the answer to *Feeling anxious*. Of the 1,503 women, 980 (65.2%) answered *Yes* and 523 (34.8%) answered *No* (Figure 4.2). The imbalance is moderate, about two to one. A model that always answered "anxious" would be right 65.2% of the time, which is the floor any useful model must clear. Because of the imbalance, the study reports balanced accuracy, recall, specificity, F1 and the Matthews correlation coefficient next to plain accuracy.

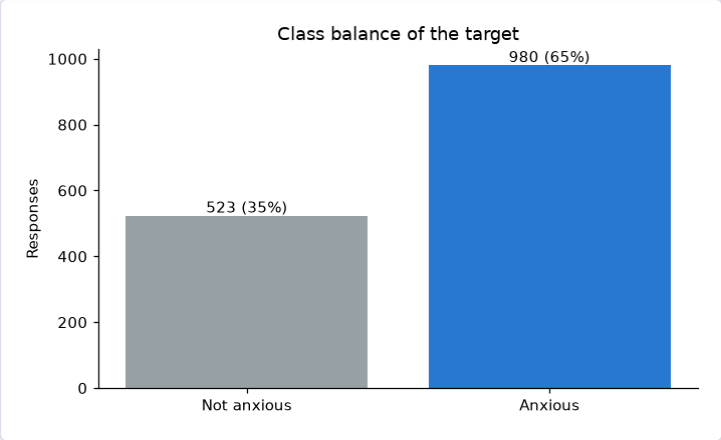


Figure 4.2. Class balance of the target.

4.5 Association of each question with the target

Cramér's V measures how strongly two categorical variables are associated, from 0 (none) to 1 (perfect). Figure 4.3 shows it for each question against the target. *Feeling of guilt* stands well above the rest. No single question comes close to determining the target, so a model has to combine answers to do well, which is why the linear baseline falls far behind the tree-based models in Chapter 6.

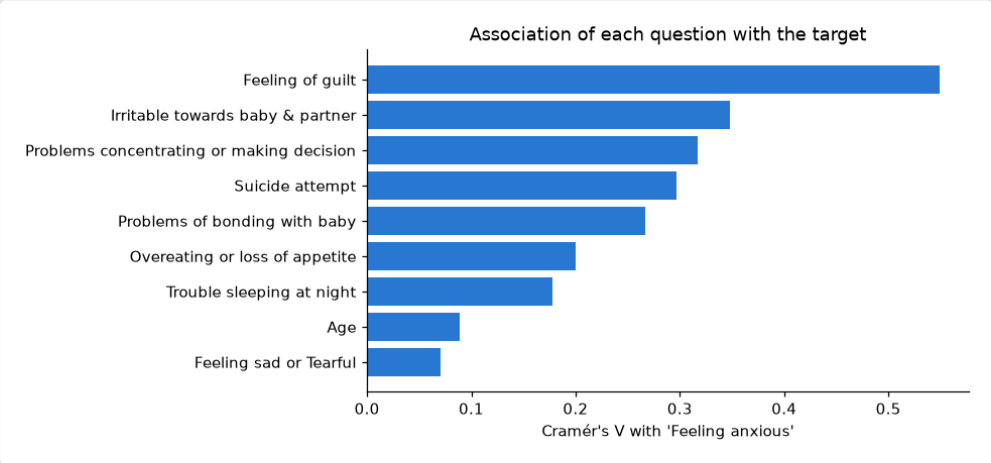


Figure 4.3. Cramér's V between each question and the target.

4.6 Repeated answer patterns

This is the most important property of the dataset. Ignoring the timestamp, the 1,503 rows contain only **326 distinct answer patterns**. The remaining 1,177 rows (78.3%) are exact repeats of a pattern that occurs earlier in the file. The most frequent pattern appears 33 times, and only 1 pattern appears exactly once, so almost every pattern in the file has at least one copy.

A second fact makes the repeats decisive: **no feature pattern occurs with both target values**. Whenever two women gave the same nine answers, they also gave the same answer to *Feeling anxious*. A model that simply memorised every training pattern together with its label would therefore be right on every test row whose pattern it had already seen. Chapter 9 measures how many test rows that is and what happens to accuracy when it is prevented.

WHY THE ROWS ARE KEPT

The study does not delete the repeated rows. Earlier papers used all 1,503 rows with a random split, so the same protocol is needed for a fair comparison. The repeats are handled by reporting a second, duplicate-aware validation next to the standard one.

4.7 Ethical considerations

The data contain self-reported mental-health answers, including a question about suicide attempt. They are public and carry no direct identifiers, but they remain sensitive. Three rules follow for this project: the raw file is never modified; results are reported only in aggregate, with individual explanations shown for anonymous test rows; and nothing in this book is a diagnostic tool or may be used to assess an individual.

CHAPTER 5

Methodology

Every step of the study, from the raw file to the final robustness check, with the reasons for each design decision.

5.1 Study design at a glance

The study is one notebook of 57 small tasks in ten parts. Figure 5.1 shows the flow. Every step that learns something from data (encoding, tuning, stacking) learns it from training data only.

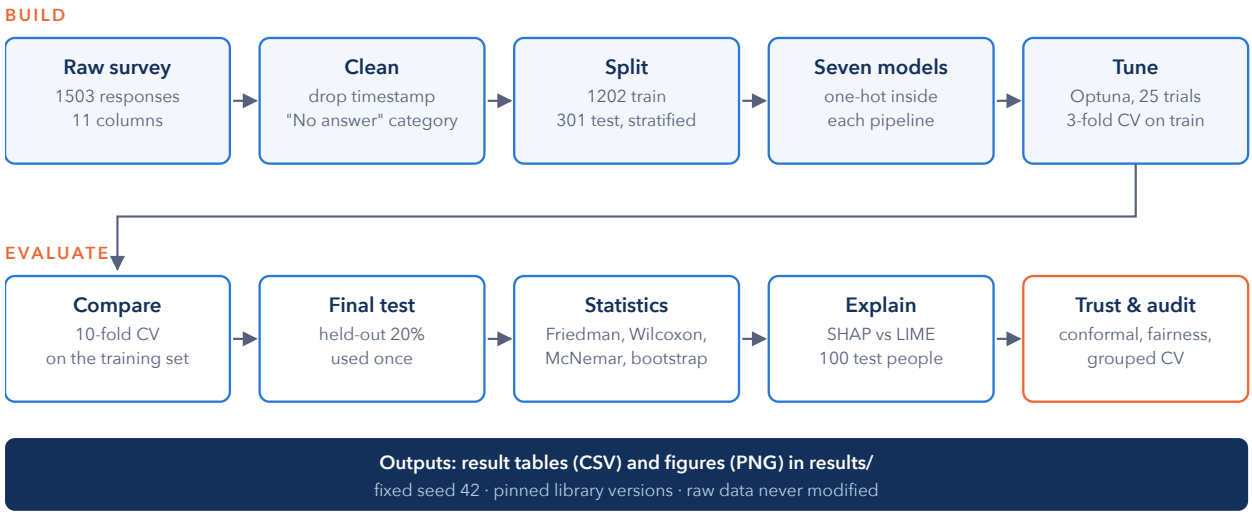


Figure 5.1. The study pipeline. The upper row builds the models; the lower row evaluates them. The last box contains the duplicate audit that the rest of the book depends on.

5.2 Data preparation

- **Cleaning.** The timestamp is dropped because it carries no clinical meaning and could leak the order of data entry. Empty cells become the category "No answer".
- **Target.** *Feeling anxious* is coded 1 for *Yes* and 0 for *No*.
- **Encoding.** Each answer is first given an integer code, which keeps the data compact and lets the explanation methods work on the original nine questions. Inside every model pipeline the codes are then **one-hot encoded**, so no model is told that "Sometimes" lies numerically between "No" and "Yes". Because the encoder is part of the pipeline, it is refitted inside each cross-validation fold and never sees validation data.
- **Split.** A stratified 80/20 split with seed 42 gives 1,202 training and 301 test people, both with 65.2% anxious answers. The test set is used once, at the end.

- **No scaling, no resampling.** All inputs are one-hot indicators, so scaling is unnecessary. The two-to-one class ratio is handled by reporting imbalance-aware metrics instead of by synthetic oversampling, which would create further near-duplicate rows.

5.3 The seven competing models

Seven models were chosen to cover the main families in the literature on this dataset, from a transparent baseline to the proposed hybrid. Table 5.1 lists their settings.

Table 5.1. The seven competing models and their settings. All use one-hot encoding inside a pipeline and random seed 42.

#	Model	Family	Settings	Why it is included
1	Logistic Regression	Linear	max_iter 2000, default L2 penalty	Transparent baseline; shows what a model without interactions can do
2	Random Forest	Bagging	300 trees	Strong, stable classic
3	XGBoost	Boosting	Tuned: 197 trees, depth 8, learning rate 0.211, subsample 0.69, column sample 0.61	Most widely used boosting model in health-ML papers
4	LightGBM	Boosting	Tuned: 285 trees, 56 leaves, learning rate 0.161, min. child samples 9, column sample 0.75	Faster, leaf-wise boosting
5	CatBoost	Boosting	400 iterations, depth 6, learning rate 0.1	Boosting designed for categorical data
6	MLP Neural Net	Neural network	Hidden layers of 64 and 32 units, up to 1000 iterations	Represents deep learning
7	**Hybrid Stack (proposed)**	**Stacking**	**Base: CatBoost, XGBoost, Random Forest, MLP. Meta: logistic regression on out-of-fold probabilities, cv = 5**	**Combines four families that make different mistakes**

5.4 The proposed Hybrid Stack

Figure 5.2 shows the architecture of the proposed model. The four base models were chosen for diversity: two boosting models with different tree-growing strategies, one bagging model and one neural network. LightGBM is left out of the stack because it is the closest relative of XGBoost and would add little diversity. The meta-learner is deliberately simple. A logistic regression on four probabilities has five parameters, so it cannot overfit the way a deep meta-learner could, and its weights can be read directly.

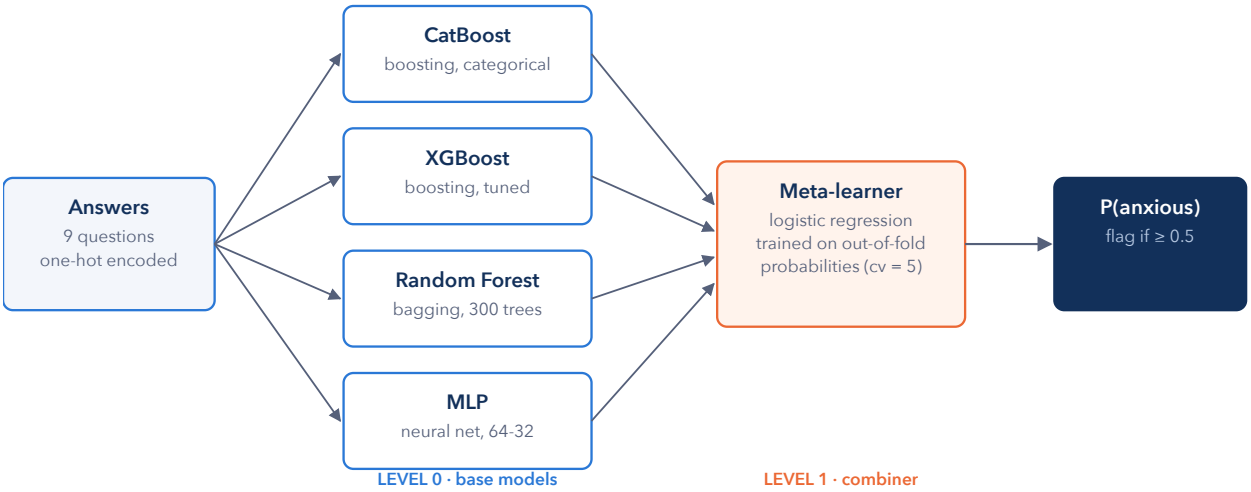


Figure 5.2. Architecture of the Hybrid Stack. Each base model outputs a probability; the meta-learner is trained only on probabilities that each base model produced for people it had not been trained on.

Training proceeds in three steps. First, the training set is cut into five folds, and each base model is trained five times, each time predicting the fold that was held back. This yields one honest probability per person per base model. Second, the logistic regression is fitted on those four columns of probabilities. Third, each base model is refitted on the whole training set for use at prediction time.

5.5 Hyper-parameter tuning

XGBoost and LightGBM are sensitive to their settings and were tuned with Optuna [22], a Bayesian optimiser that proposes each new setting from the results of the earlier ones (a tree-structured Parzen estimator, seeded for reproducibility). Each model received 25 trials, and each trial was scored by three-fold cross-validation on the training set only. Table 5.2 gives the search ranges. The other models use fixed, conventional settings, which keeps the search modest and the risk of overfitting the validation folds low.

Table 5.2. Hyper-parameter search space and the values selected.

Model	Hyper-parameter	Search range	Selected
XGBoost	Number of trees	100 – 500	197
	Maximum depth	3 – 10	8
	Learning rate	0.02 – 0.3 (log scale)	0.211
	Row subsample	0.6 – 1.0	0.687
	Column subsample per tree	0.6 – 1.0	0.611
LightGBM	Number of trees	100 – 500	285
	Number of leaves	8 – 64	56
	Learning rate	0.02 – 0.3 (log scale)	0.161
	Minimum samples per leaf	1 – 20	9
	Column subsample per tree	0.6 – 1.0	0.752

5.6 Validation protocol

Figure 5.3 shows how the data are used. Three layers of validation keep model selection apart from the final measurement.

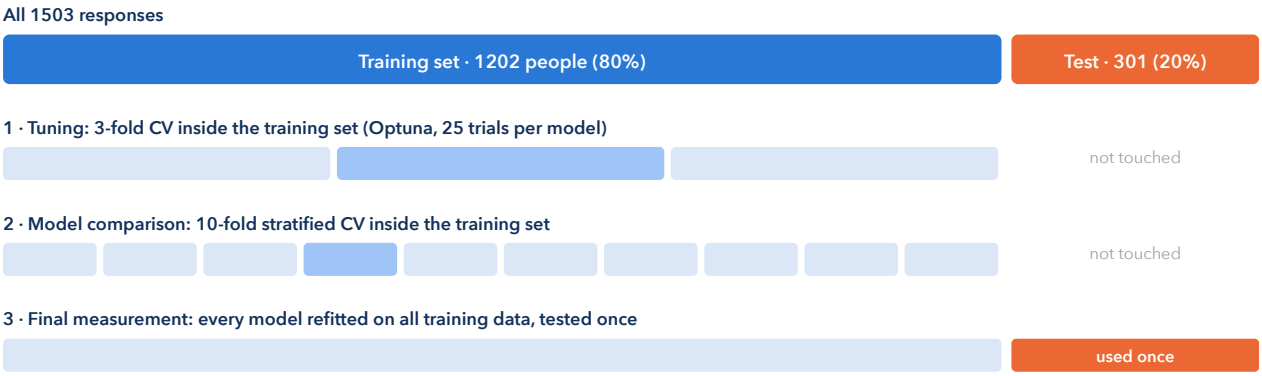


Figure 5.3. Validation protocol. The darker fold in each row is the one held out in that round. The test set stays untouched until the last step.

- **Tuning** uses three-fold cross-validation on the training set.
- **Comparison** uses ten-fold stratified cross-validation on the training set. Ten accuracies per model give a mean, a standard deviation and paired samples for the statistical tests.
- **Final measurement** refits each model on the full training set and evaluates it once on the 301 test people.

5.7 Performance metrics

With TP, TN, FP and FN the counts of true positives, true negatives, false positives and false negatives, the study reports the following.

Table 5.3. Performance metrics used in the study.

Metric	Definition	What it tells the reader
Accuracy	(TP + TN) / all	Share of all people classified correctly
Balanced accuracy	(recall + specificity) / 2	Accuracy that is not inflated by the larger class
Precision	TP / (TP + FP)	Of those flagged anxious, how many are
Recall (sensitivity)	TP / (TP + FN)	Of the anxious women, how many are found; the key number for screening
Specificity	TN / (TN + FP)	Of the not-anxious women, how many are correctly cleared

Metric	Definition	What it tells the reader
F1	Harmonic mean of precision and recall	One number balancing the two
MCC	Correlation between prediction and truth, from −1 to 1	Robust summary under class imbalance
ROC-AUC	Area under the ROC curve	How well probabilities rank anxious above not-anxious, at every threshold
Brier score	Mean of (probability − outcome)^2	Quality of the probabilities themselves; lower is better

5.8 Statistical testing

Four procedures test whether observed differences are real. They follow the recommendations of Demšar for comparing several classifiers [26].

- **Friedman test** on the ten fold accuracies of all seven models: is there any difference among the models at all?
- **Wilcoxon signed-rank tests**, Hybrid Stack against each other model on the same ten folds, with **Holm correction** for the six comparisons.
- **McNemar test** (exact binomial form) on the test set: of the people on whom two models disagree, does one model win more often than chance would allow?
- **Bootstrap confidence interval**: the 301 test outcomes are resampled 2,000 times to give a 95% interval for the hybrid's test accuracy.

5.9 Explainability protocol

The Hybrid Stack is explained for the first 100 test people. SHAP is computed exactly against a background of 30 randomly chosen training people. LIME uses 1,000 perturbed samples per person and treats all features as categorical. The two are then compared on four criteria.

Table 5.4. The four criteria on which SHAP and LIME are compared.

Criterion	How it is measured	Better when
Agreement	Spearman rank correlation of global importance; mean per-person rank correlation; share of people with the same top question; overlap of the top three; share of contributions with the same sign	Higher: the two tell the same story
Faithfulness	Deletion test: replace a person's top 1, 2 or 3 questions (as ranked by the method) with the answers of the 30 background people and record the mean absolute change in predicted probability. Random choice of questions is the reference	Higher: the method found the answers that really move the model
Stability	Repeat the explanation for 20 people with different background people (SHAP) or a different random seed (LIME); Spearman correlation between the two runs	Higher: the same person gets the same explanation twice
Cost	Seconds per explained person	Lower

As a cross-check that does not depend on either method, **permutation importance** is computed for all seven models: one question at a time is shuffled in the test set and the loss of accuracy is recorded.

5.10 Trustworthy-AI checks

- **Calibration.** A reliability curve with eight quantile bins and the Brier score.
- **Conformal prediction.** Out-of-fold probabilities of the hybrid on the training set give the calibration scores, $s = 1 - p(\text{true class})$. For a target error rate α, the threshold q is the $\lceil (n+1)(1-\alpha) \rceil / n$ quantile of those scores, and a class enters a test person's prediction set if $1 - p(\text{class}) \le q$. Targets of 90%, 95% and 99% are evaluated.
- **Counterfactual.** For one test person flagged anxious, an exhaustive search over all single-answer changes (and pairs, if no single change suffices) finds every change that brings the probability below 0.5. Age is excluded because it cannot be changed.
- **Subgroup check.** Test accuracy of the hybrid in each of the five age bands.

5.11 Duplicate-aware robustness check

The last part of the study addresses the repeated answer patterns of Section 4.6 in two ways. First, the test set is divided into people whose exact pattern also occurs in the training set ("twins") and people whose pattern does not, and accuracy is computed separately. Second, all seven models are re-evaluated with **grouped five-fold cross-validation** on the full dataset: every answer pattern is a group, and each group falls entirely into one fold. Figure 5.4 illustrates the difference.

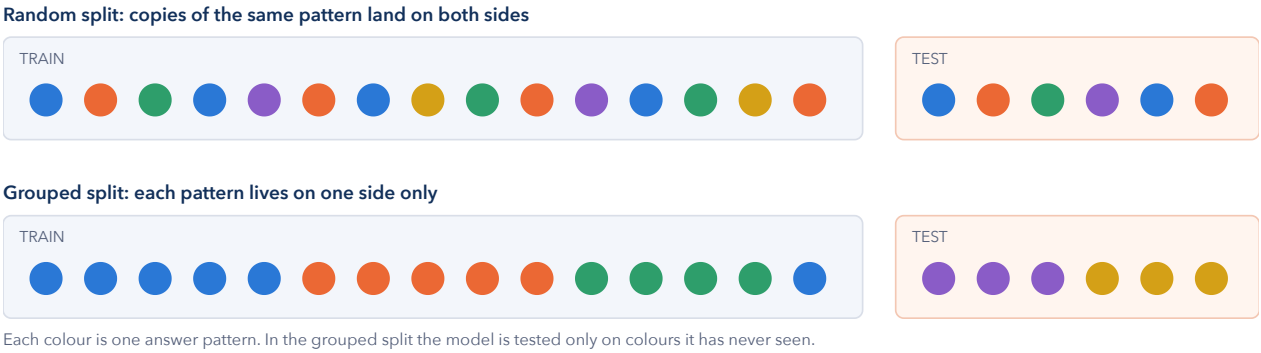


Figure 5.4. Random versus grouped splitting when a dataset contains repeated answer patterns.

5.12 Software and reproducibility

The study runs on Python 3.11.16 with the library versions in Table 5.5. All random processes use seed 42. The complete run takes a few minutes on a laptop. Appendix A gives the exact commands.

Table 5.5. Pinned library versions.

Library	Version	Library	Version	Library	Version
numpy	2.4.6	pandas	3.1.1	scipy	1.17.1
scikit-learn	1.9.1	matplotlib	3.11.2	xgboost	3.2.0
lightgbm	4.7.0	catboost	1.2.10	optuna	5.0.0
shap	0.51.0	lime	0.2.0.1		

CHAPTER 6

Results I: The Model Competition

Seven models, one protocol: who wins, by how much, and whether the margin means anything.

6.1 Hyper-parameter search

Figure 6.1 shows the best three-fold accuracy found by Optuna after each trial. Both searches improve quickly and then flatten, which indicates that 25 trials were enough for this small problem. The best settings reached 95.51% (XGBoost) and 95.76% (LightGBM) in three-fold cross-validation. These tuning scores are lower than the ten-fold scores reported next because each three-fold model is trained on only two thirds of the training set.

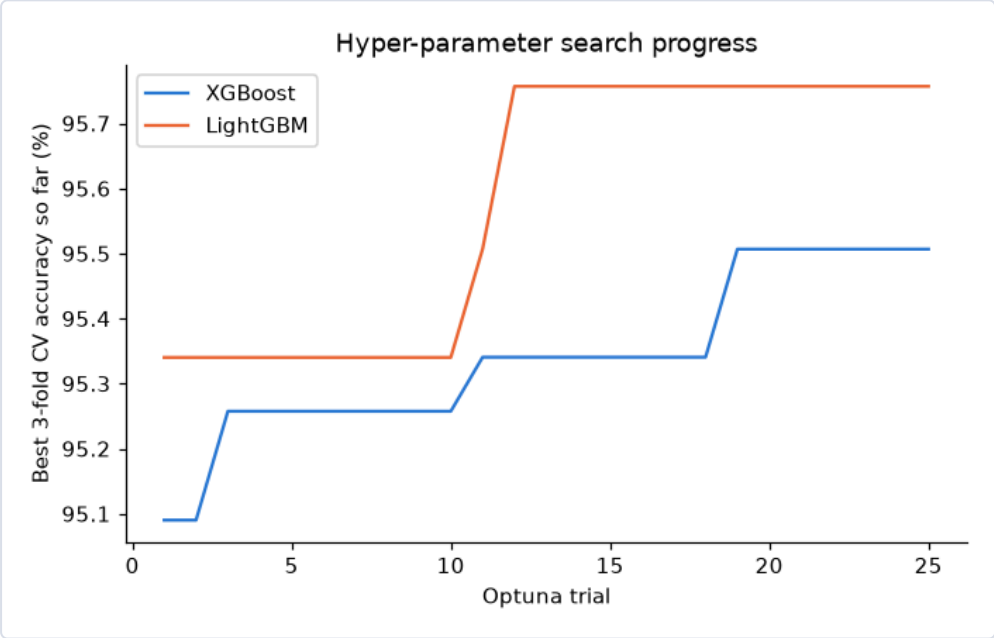


Figure 6.1. Best three-fold cross-validation accuracy after each Optuna trial.

6.2 Ten-fold cross-validation

Table 6.1 and Figure 6.2 give the ten-fold cross-validation results on the training set. Six models lie between 97.59% and 98.25%, a spread of 0.66 points that is smaller than the fold-to-fold standard deviation of any one of them. Logistic regression is far behind at 82.70%, which confirms that the target depends on combinations of answers that a linear model cannot represent.

Table 6.1. Ten-fold stratified cross-validation accuracy on the training set.

Model	Mean accuracy	Std. deviation
CatBoost	98.25%	± 1.44
LightGBM	98.08%	± 1.47
Hybrid Stack (proposed)	**98.00%**	**± 1.26**
Random Forest	97.92%	± 1.37
XGBoost	97.83%	± 1.53
MLP Neural Net	97.59%	± 1.39
Logistic Regression	82.70%	± 2.34

Bars start at 75% to make the differences among the top six visible.

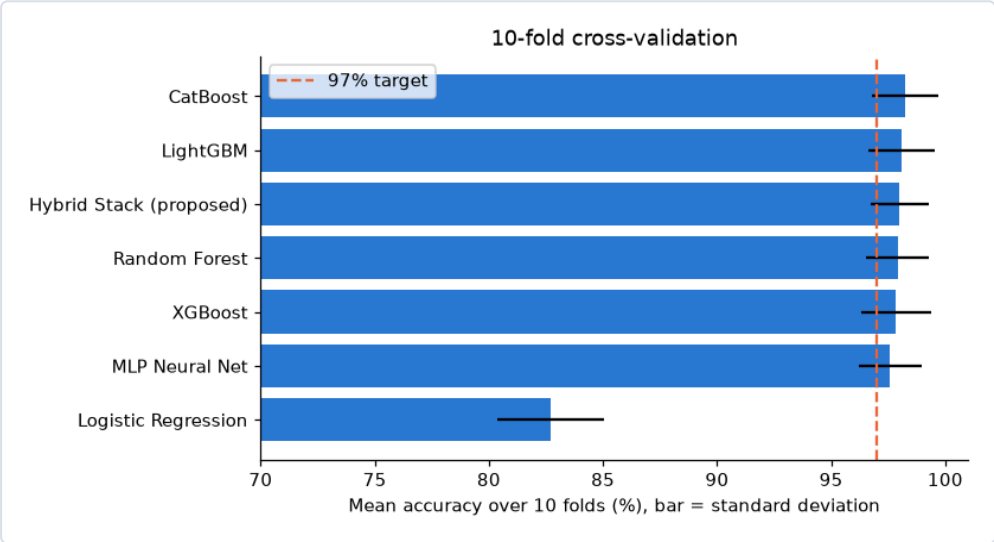


Figure 6.2. Mean ten-fold cross-validation accuracy; the error bar is the standard deviation across folds.

6.3 Held-out test set

Each model was then refitted on the full training set and evaluated once on the 301 held-out people (196 anxious, 105 not anxious). Table 6.2 gives all nine metrics.

Table 6.2. Test-set performance of the seven models. Accuracy, balanced accuracy, precision, recall, specificity and F1 in percent.

Model	Acc.	Bal. acc.	Prec.	Recall	Spec.	F1	MCC	AUC	Brier
CatBoost	100.00	100.00	100.00	100.00	100.00	100.00	1.000	1.000	0.002
Random Forest	99.34	99.49	100.00	98.98	100.00	99.49	0.986	1.000	0.014
MLP Neural Net	98.67	98.98	100.00	97.96	100.00	98.97	0.971	0.997	0.015
Hybrid Stack (proposed)	**98.67**	**98.98**	**100.00**	**97.96**	**100.00**	**98.97**	**0.971**	**1.000**	**0.006**
LightGBM	97.67	98.21	100.00	96.43	100.00	98.18	0.951	1.000	0.022
XGBoost	97.01	97.70	100.00	95.41	100.00	97.65	0.937	0.995	0.020
Logistic Regression	81.73	78.67	84.06	88.78	68.57	86.35	0.590	0.895	0.122

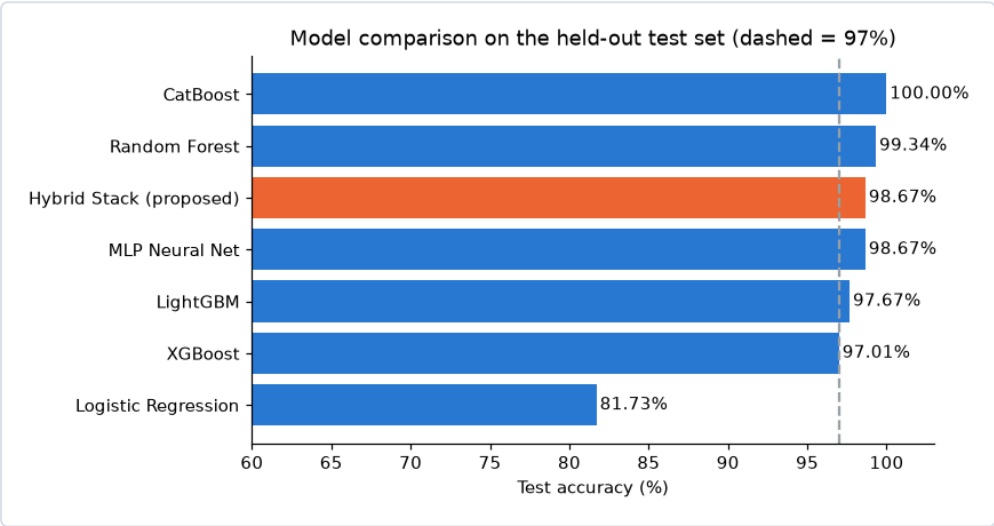


Figure 6.3. Test accuracy of every model. The proposed Hybrid Stack is shown in orange; the dashed line marks 97%.

Three points stand out.

- **6 of the seven models exceed 97%.** CatBoost classifies all 301 test people correctly, the random forest misses 2, and the Hybrid Stack and the MLP each miss 4.
- **The Hybrid Stack reaches 98.67%** with a 95% bootstrap confidence interval of 97.34–99.67%. The interval is more than two points wide, so test accuracies that differ by a point or less cannot be told apart on a test set of this size.
- **The hybrid's probabilities are among the best.** Its Brier score of 0.006 is second only to CatBoost's 0.002 and less than half that of the random forest (0.014), although the forest has slightly higher accuracy. The meta-learner sharpens the probabilities of its base models.

CATBOOST SCORES 100%. IS THAT A PROBLEM?

On its own it is not an error in the code: the split, the pipeline and the test are clean. It is, however, not evidence of a perfect model. Chapter 9 shows that 93.4% of the test people have an exact copy of their answers in the training set, and that CatBoost's accuracy on unseen patterns is 85.2%. The honest headline for CatBoost is its cross-validation result, 98.25% ± 1.44, reported together with the grouped figure.

6.4 Overfitting check

Figure 6.4 compares training and test accuracy. All six non-linear models reach 100.00% on the training set, and their test accuracy is at most 2.99 points lower (Table 6.3). A gap this small would normally indicate little overfitting. Here it must be read with the duplicates in mind: the test set looks like the training set because much of it *is* the training set. The grouped validation in Chapter 9 is the proper overfitting check.

Table 6.3. Training accuracy, test accuracy and the gap between them (percentage points).

Model	Train accuracy	Test accuracy	Gap
CatBoost	100.00%	100.00%	0.00
Random Forest	100.00%	99.34%	0.66
MLP Neural Net	100.00%	98.67%	1.33
Hybrid Stack (proposed)	**100.00%**	**98.67%**	**1.33**
LightGBM	100.00%	97.67%	2.33
XGBoost	100.00%	97.01%	2.99
Logistic Regression	83.44%	81.73%	1.72

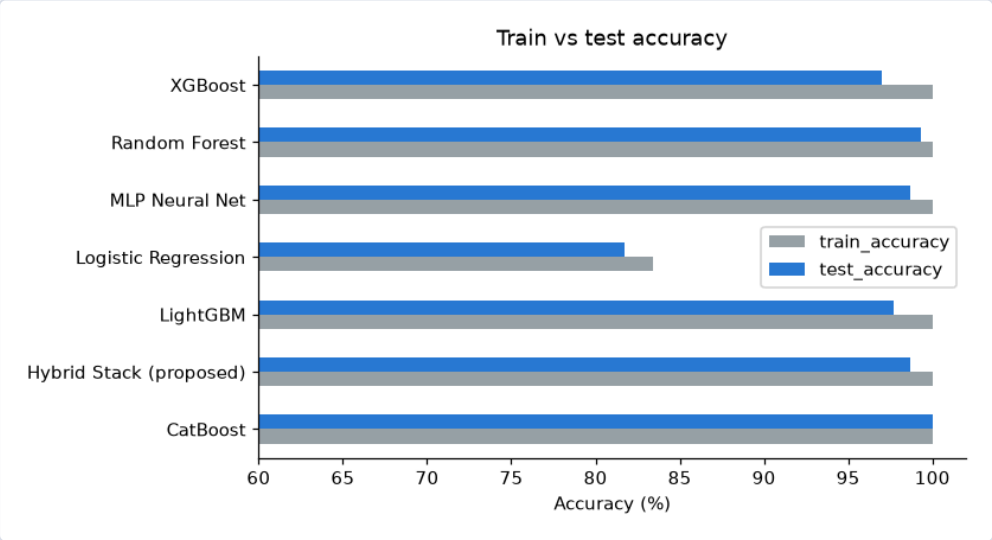


Figure 6.4. Training accuracy (grey) and test accuracy (blue) of every model.

6.5 ROC curves, confusion matrices and calibration

The ROC curves (Figure 6.5) of the six non-linear models hug the top-left corner, with areas between 0.995 and 1.000. Logistic regression reaches 0.895.

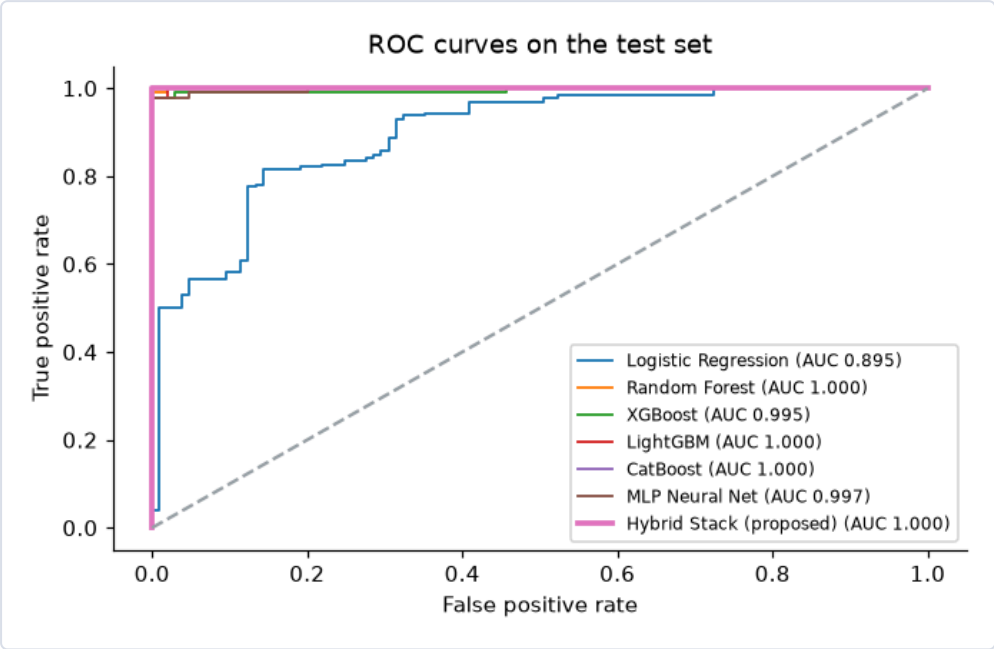


Figure 6.5. ROC curves on the test set. The Hybrid Stack is drawn with a thicker line.

Table 6.4 gives the confusion matrix of every model, and Figure 6.6 shows the hybrid's. A notable pattern is that **none of the six non-linear models produces a single false positive**: every error is a missed anxious woman. The Hybrid Stack misses 4 of 196 (recall 97.96%). For a screening tool this is the less desirable kind of error, and Section 11.3 returns to it: the decision threshold of 0.5 could be lowered to trade a few false alarms for fewer misses.

Table 6.4. Confusion-matrix counts on the 301 test people.

Model	True negative	False positive	False negative	True positive	Errors
CatBoost	105	0	0	196	0
Random Forest	105	0	2	194	2
MLP Neural Net	105	0	4	192	4
Hybrid Stack (proposed)	**105**	**0**	**4**	**192**	**4**
LightGBM	105	0	7	189	7
XGBoost	105	0	9	187	9
Logistic Regression	72	33	22	174	55

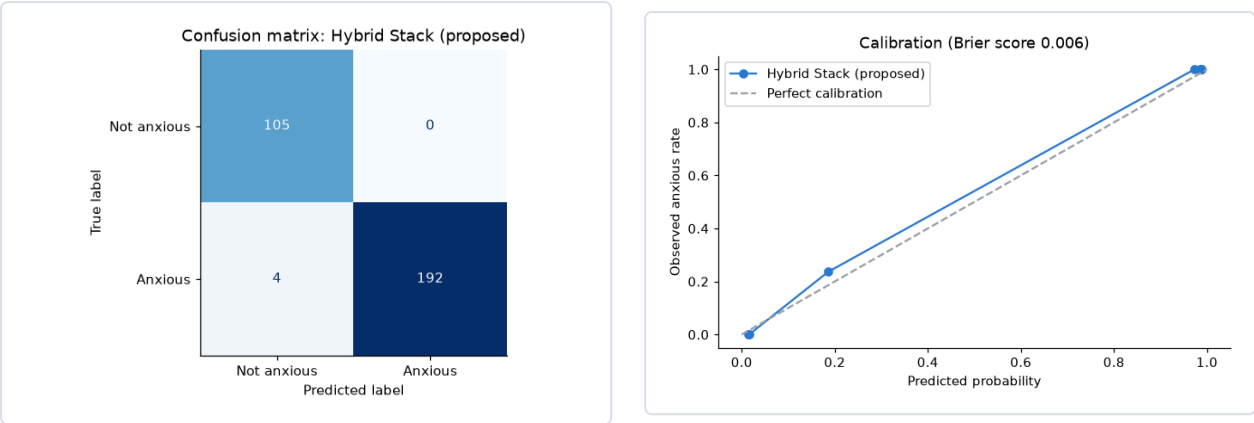


Figure 6.6. Left: confusion matrix of the Hybrid Stack. Right: its calibration curve; points on the dashed diagonal mean that predicted probabilities match observed rates.

The calibration curve lies close to the diagonal. Most predictions sit at the two ends, near 0 and near 1, which reflects how confidently the model separates the classes on this data.

6.6 Are the differences statistically real?

The Friedman test on the ten cross-validation folds rejects the hypothesis that all seven models perform equally ($\chi^2 = 37.68$, $p < 0.001$). The pairwise tests in Table 6.5 show where the difference lies.

Table 6.5. Hybrid Stack against each other model. Left: Wilcoxon signed-rank test on ten cross-validation folds with Holm correction. Right: exact McNemar test on the test set.

Hybrid Stack versus	Mean diff. (points)	Wilcoxon p	Holm p	Only hybrid right	Only other right	McNemar p
Logistic Regression	+15.31	0.002	0.012	53	2	< 0.001
CatBoost	-0.25	0.375	1.000	0	4	0.125
MLP Neural Net	+0.41	0.375	1.000	0	0	1.000
XGBoost	+0.17	0.500	1.000	5	0	0.062
LightGBM	-0.08	0.750	1.000	3	0	0.250
Random Forest	+0.08	1.000	1.000	0	2	0.500

The Hybrid Stack is significantly better than logistic regression on both tests (Holm-corrected $p = 0.012$; McNemar $p < 0.001$). Against the other five models no difference is significant: every Holm-corrected p-value is 1.0, and the smallest McNemar p-value is 0.062. On the test set the hybrid and the MLP make exactly the same errors.

ANSWER TO RQ1

CatBoost has the highest cross-validation and test accuracy, but the six non-linear models, including the proposed Hybrid Stack, cannot be separated statistically on this dataset. The Friedman test is significant only because logistic regression is far behind. The honest claim for the Hybrid Stack is therefore "as accurate as the best single models, with better-calibrated probabilities than most of them", not "more accurate".

CHAPTER 7

Results II: Explaining the Hybrid Stack

Which answers drive the model, whether two explanation methods tell the same story, and which of them deserves more trust.

7.1 What SHAP says

Averaging the absolute SHAP values over the 100 explained test people gives the global importance in Figure 7.1. *Feeling of guilt* accounts for 37.3% of the total, about 2.7 times the share of the next question. It is followed by *Irritable towards baby & partner* (13.9%) and *Problems concentrating or making decision* (12.3%).

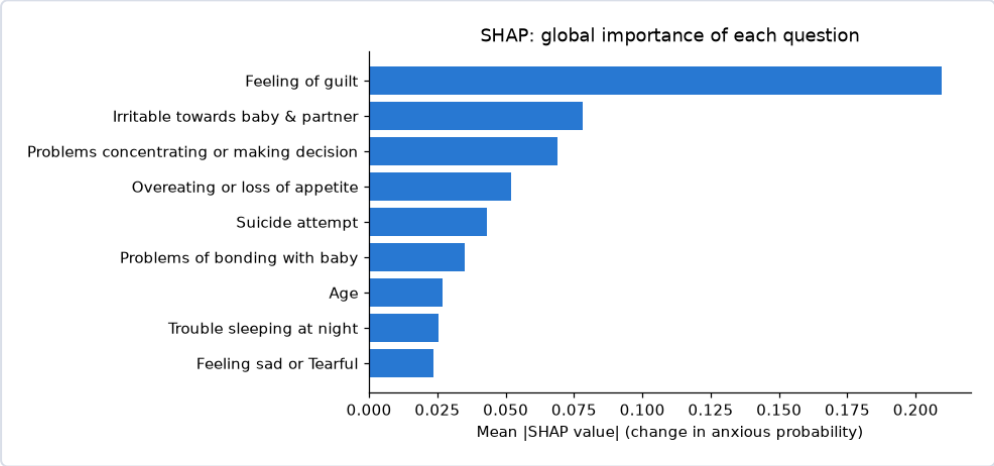


Figure 7.1. SHAP global importance: mean absolute contribution of each question to the predicted probability of being anxious.

Figure 7.2 looks inside the top question: the mean SHAP value for each answer. An answer above zero pushes the prediction towards "anxious". The direction mirrors the raw data of Section 4.2, including its counter-intuitive ordering, which shows that the model has learned the dataset faithfully and that an explanation can only be as meaningful as the data behind it.

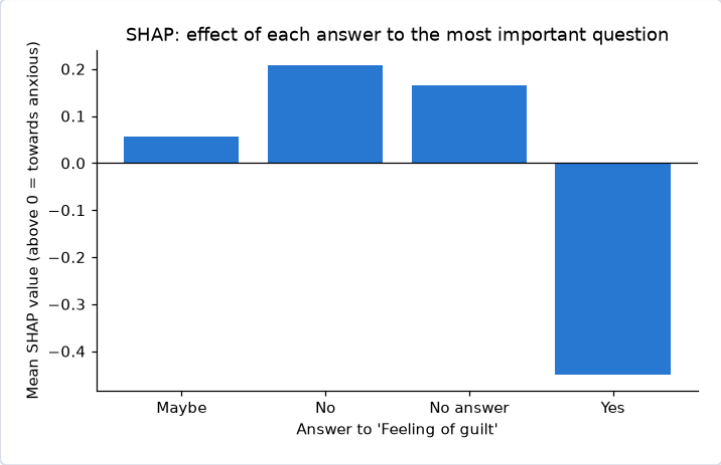


Figure 7.2. Mean SHAP value of each answer to the most important question.

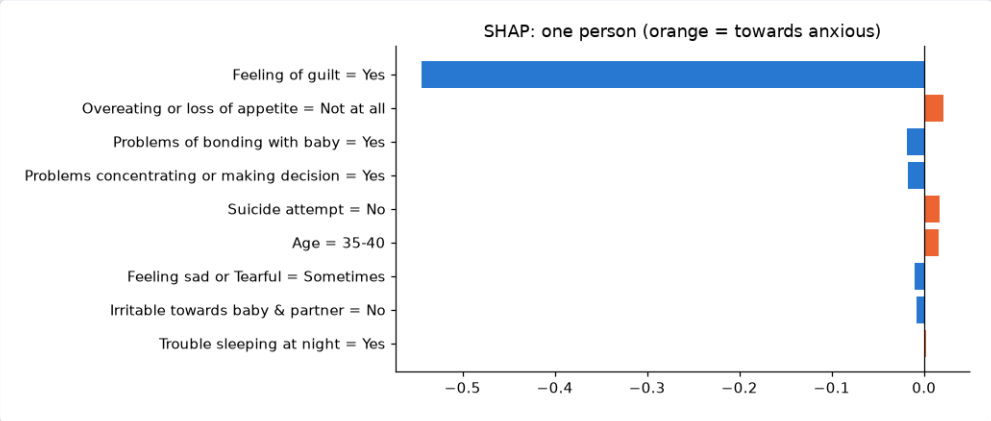


Figure 7.3. SHAP explanation of a single test person. Orange bars push towards "anxious", blue bars away from it.

Figure 7.3 explains one person. The model gives her a probability of 0.024 of being anxious (her true answer is No). The SHAP values add up exactly: the baseline of 0.566 plus contributions of -0.542 gives 0.024. This additivity is what allows a clinician to read the bars as a complete account of the prediction.

7.2 What LIME says

LIME's global ranking (Figure 7.4) also puts *Feeling of guilt* first, with 38.1% of the total. Figure 7.5 explains the same person as before. The largest bars match SHAP's; the smaller ones differ in size and occasionally in order.

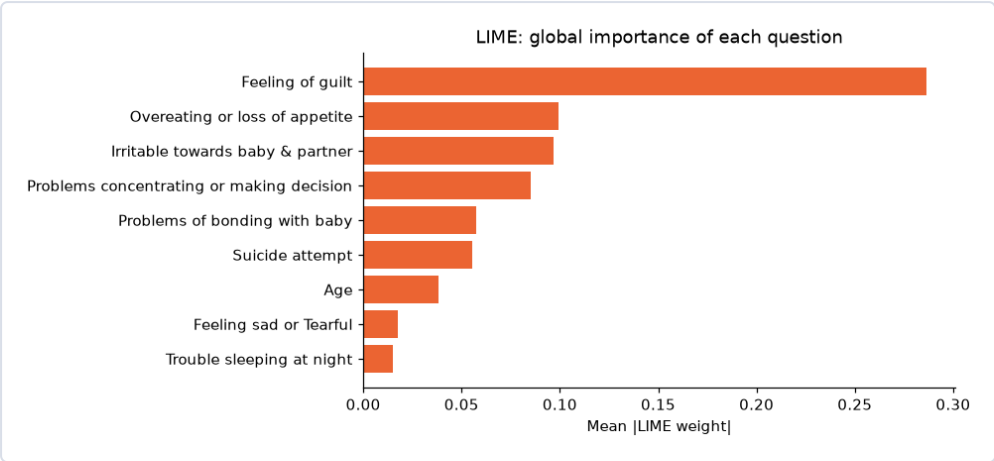


Figure 7.4. LIME global importance: mean absolute weight of each question.

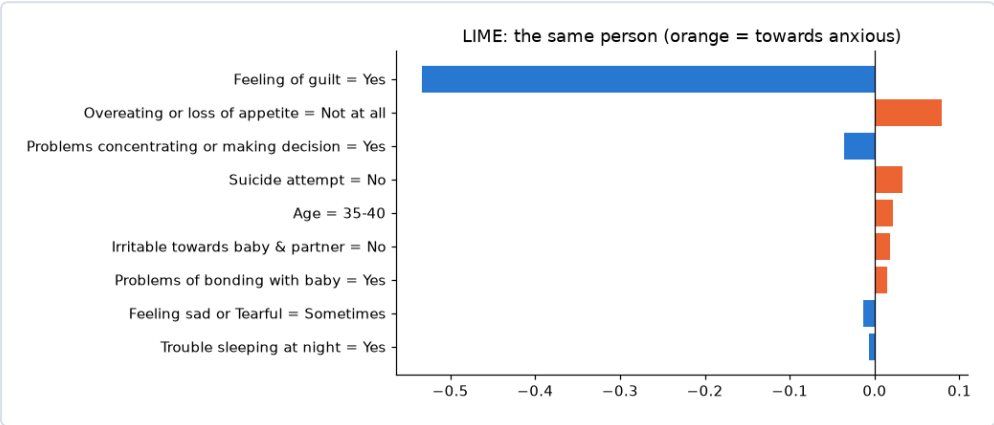


Figure 7.5. LIME explanation of the same person as in Figure 7.3.

7.3 Do SHAP and LIME agree?

Table 7.1 and Figure 7.6 put the two global rankings next to each other, and Table 7.2 measures their agreement.

Table 7.1. Share of total importance assigned to each question by SHAP and by LIME (percent).

Question	SHAP	LIME
Feeling of guilt	37.3	38.1
Irritable towards baby & partner	13.9	12.9
Problems concentrating or making decision	12.3	11.3
Overeating or loss of appetite	9.2	13.2
Suicide attempt	7.6	7.4
Problems of bonding with baby	6.2	7.6
Age	4.8	5.1
Trouble sleeping at night	4.5	2.0
Feeling sad or Tearful	4.2	2.3

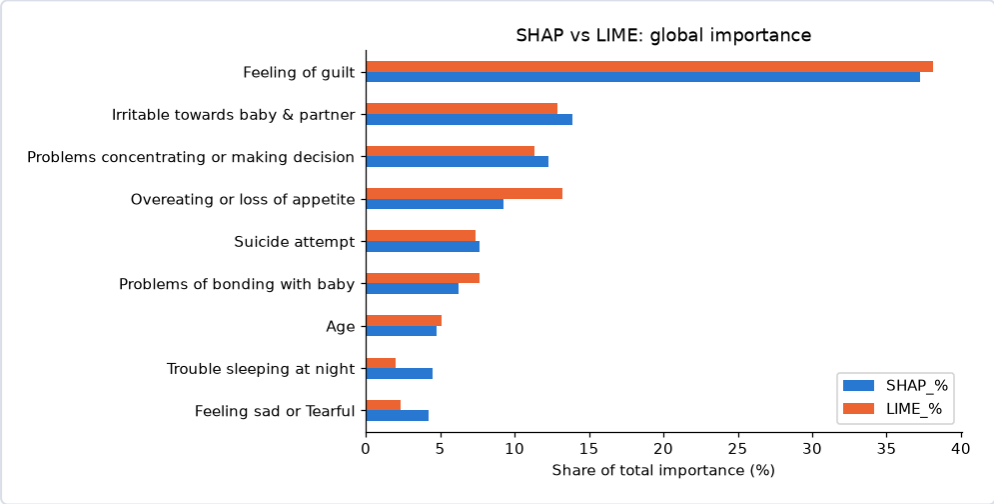


Figure 7.6. SHAP and LIME global importance side by side.

Table 7.2. Agreement between SHAP and LIME over 100 test people.

Measure	Value	Reading
Spearman correlation of the global rankings	0.917	Very strong: the same overall picture
Mean per-person Spearman correlation	0.534	Moderate: individual explanations differ
Same top question	71.0%	For about 3 people in 10 the two methods name a different main reason
Overlap of the top three questions	64.7%	On average about two of the three match
Contributions with the same sign	77.6%	About one contribution in 4 points the opposite way

The two methods agree on the big picture and differ in the detail. A reader who only saw the global bar charts would conclude that SHAP and LIME are interchangeable. At the level where a clinician would use them, one woman at a time, they name a different main reason for roughly 29% of people. This is the disagreement problem [16] made concrete, and it is why the next section asks which method is right more often.

7.4 Which explanation is more faithful?

Faithfulness is tested by deletion. If an explanation has found the answers that really drive a prediction, replacing those answers should change the prediction a lot. Table 7.3 gives the mean absolute change in predicted probability after replacing the top one, two and three questions named by each method, with random questions as the reference; Figure 7.7 shows the same numbers.

Table 7.3. Deletion-based faithfulness: mean absolute change in predicted probability after replacing the top-ranked questions. Higher is better.

Questions replaced	SHAP	LIME	Random	SHAP over LIME
Top 1	0.301	0.237	0.098	+27%
Top 2	0.414	0.329	0.181	+26%
Top 3	0.471	0.387	0.241	+22%

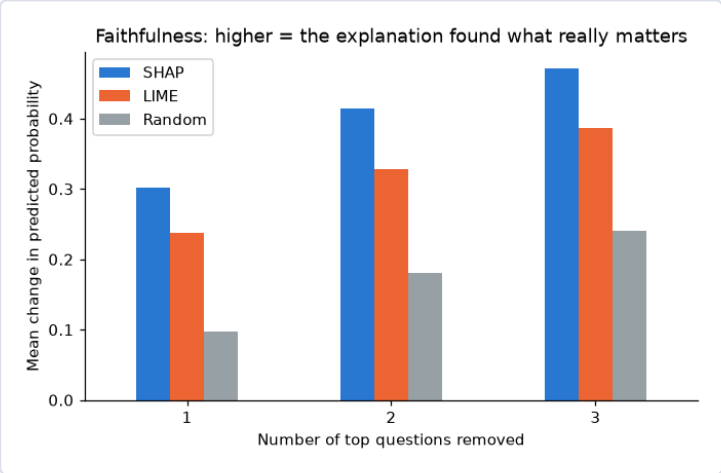


Figure 7.7. Faithfulness of SHAP, LIME and a random ranking.

Both methods are far better than chance: replacing a single question picked by SHAP moves the prediction by 0.301, about 3 times the effect of a random question. SHAP is ahead of LIME at every depth, by 27% for the top question and 22% for the top three. When the two disagree about a person, SHAP's ranking is the one that better reflects what the model actually does.

7.5 Stability and cost

Table 7.4. SHAP versus LIME: summary of all criteria.

Criterion	SHAP	LIME
Most important question	Feeling of guilt (37.3%)	Feeling of guilt (38.1%)
Top three questions	Feeling of guilt, Irritable towards baby & partner, Problems concentrating or making decision	Feeling of guilt, Overeating or loss of appetite, Irritable towards baby & partner
Faithfulness, top 1 replaced	0.301	0.237
Faithfulness, top 3 replaced	0.471	0.387
Stability (Spearman between two runs)	0.908	0.903
Seconds per person	0.14	0.03
Contributions add up to the prediction	Yes	No

Timings depend on the machine and vary from run to run; their ratio is the meaningful quantity.

The two methods are about equally stable: repeating the explanation with different background people (SHAP) or a different random seed (LIME) gives rank correlations of 0.908 and 0.903. LIME is roughly 5 times faster. With nine features both are quick enough for interactive use, so speed is not a reason to prefer LIME here.

> **ANSWER TO RQ2 AND RQ3**
>
> *Feeling of guilt* is the dominant driver of the model's predictions according to both methods. SHAP ranks *Irritable towards baby & partner*, *Problems concentrating or making decision* and *Overeating or loss of appetite* next. SHAP and LIME agree on this global picture but give a different main reason for about 29% of individuals. SHAP is the more faithful of the two, equally stable, slower but still fast, and the only one whose contributions sum to the prediction. For per-person explanations of this model, SHAP is the method to use.

7.6 Do the seven models agree on what matters?

Permutation importance offers a check that is independent of both SHAP and LIME and can be computed for every model. Table 7.5 gives the accuracy lost, in percentage points, when one question is shuffled in the test set.

Table 7.5. Permutation importance by model: loss of test accuracy (percentage points) when one question is shuffled. Darker cells mean larger loss.

Question	LR	RF	XGB	LGBM	CatB	MLP	Hybrid
Feeling of guilt	7.7	20.5	18.1	19.5	20.7	19.6	19.5
Problems concentrating or making decision	2.3	6.0	12.3	11.5	8.4	11.0	7.3
Overeating or loss of appetite	3.3	6.6	8.0	8.0	5.2	11.2	5.9
Irritable towards baby & partner	4.8	6.4	4.9	4.8	5.8	7.0	4.9
Trouble sleeping at night	1.0	3.5	3.3	2.1	4.4	4.7	3.8
Suicide attempt	1.3	4.2	3.1	4.1	4.2	6.9	2.9
Age	-0.1	2.1	4.2	4.5	3.6	7.7	2.5
Feeling sad or Tearful	1.0	3.1	2.9	1.3	3.1	7.0	2.2
Problems of bonding with baby	1.9	1.7	1.1	1.2	2.2	8.6	1.1

LR = logistic regression, RF = random forest, XGB = XGBoost, LGBM = LightGBM, CatB = CatBoost.

All seven models, including the linear baseline, lose most accuracy when *Feeling of guilt* is shuffled: about 20 points for the hybrid. The models agree less on the order of the remaining questions, and the MLP spreads its reliance more evenly than the tree models. The agreement on the top question across seven different models and three different importance methods is strong evidence that its dominance is a property of the data and not an artefact of one model or one explanation technique.

CHAPTER 8

Results III: Trustworthy-AI Checks

Three additions that a screening tool needs beyond accuracy and explanation: knowing when it is unsure, saying what would change its mind, and treating subgroups alike.

8.1 Conformal prediction

Table 8.1 and Figure 8.1 give the conformal prediction results for three coverage targets. In every case the coverage achieved on the test set is at or above the coverage promised.

Table 8.1. Conformal prediction with the Hybrid Stack on the 301 test people.

Target coverage	Actual coverage	Mean set size	One answer	Both answers	Empty set
90%	94.02%	0.94	94.02%	0.00%	5.98%
95%	97.01%	0.97	97.01%	0.00%	2.99%
99%	100.00%	1.04	96.35%	3.65%	0.00%

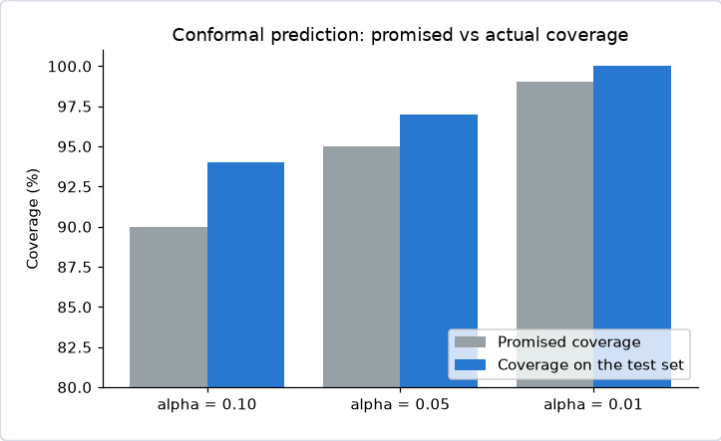


Figure 8.1. Promised and achieved coverage of the conformal prediction sets.

At the 95% target the model commits to a single answer for 97.0% of the test people and returns an empty set for the remaining 3.0%. Every single-answer set contains the true answer, so all of the model's test errors fall among the people for whom it declined to commit. An empty set arises because the model is confident so often that the calibrated threshold becomes strict; a prediction with a probability near 0.5 then fails the threshold for both classes. In practice an empty set and a two-answer set mean the same thing to a clinic: this case goes to a human. At the 99% target the behaviour changes as theory predicts: the empty sets disappear and 3.6% of people receive both answers.

WHAT THIS ADDS

Without conformal prediction the model answers all 301 people and is wrong on 4 of them with no warning. With it, at the 95% setting, the model answers about 292 people and refers about 9 to a human, and the cases it refers include the ones it would have got wrong.

8.2 A counterfactual example

For one test person whom the model flags as anxious with probability 0.987, the search of Section 5.10 found that a single changed answer is enough to reverse the decision (Table 8.2).

Table 8.2. Counterfactuals for one test person predicted anxious with probability 0.987.

Answers changed	Change	New probability
1	Irritable towards baby & partner: Yes → Sometimes	0.132
1	Irritable towards baby & partner: Yes → No	0.208

The model's decision for this woman hinges on one answer. That is useful to know in two ways. For the clinician it identifies the symptom to ask about first. For the model's developer it is a warning about sensitivity: a prediction that moves from 0.99 to 0.13 when one answer shifts by one step rests on a narrow basis, and a mis-clicked answer on a form could flip the outcome. A counterfactual describes the model, not the woman: it does not say that reducing irritability would relieve her anxiety.

8.3 Accuracy by age group

Table 8.3 and Figure 8.2 give the hybrid's test accuracy in each age band.

Table 8.3. Test accuracy of the Hybrid Stack by age group.

Age group	Test people	Accuracy	Errors
25-30	39	94.87%	2
30-35	78	100.00%	0
35-40	68	100.00%	0
40-45	69	97.10%	2
45-50	47	100.00%	0

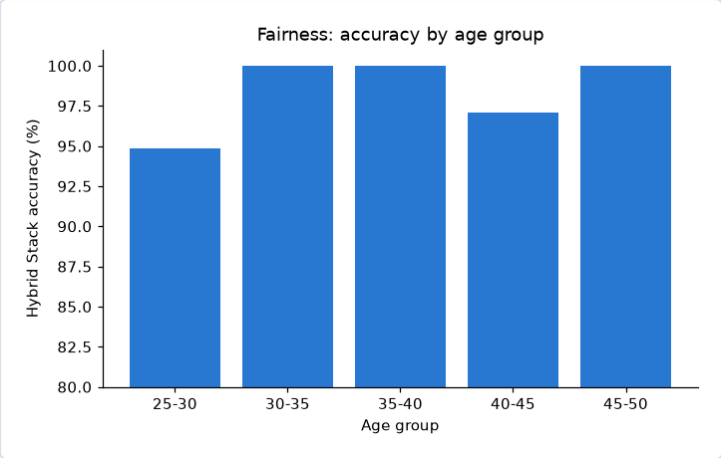


Figure 8.2. Hybrid Stack accuracy in each age group.

Accuracy ranges from 94.87% to 100.00%. The lowest value belongs to the smallest group, where it corresponds to only 2 misclassified people out of 39. With groups of 39 to 78 people, differences of this size are within what chance alone would produce, so the check finds no evidence of an age-related gap. It also cannot rule one out. Age is the only demographic variable in the data, so fairness with respect to income, education, ethnicity or region cannot be examined at all.

ANSWER TO RQ4

Yes, within the limits of the data. The hybrid's probabilities are well calibrated (Brier 0.006), its conformal sets meet their coverage promise at all three levels, a one-answer counterfactual exists for the example person, and accuracy does not differ detectably between age groups.

CHAPTER 9

Results IV: The Duplicate Audit

The check that changes how every accuracy in this book, and in earlier papers on this dataset, should be read.

9.1 Twins in the test set

Section 4.6 showed that the 1,503 rows contain only 326 distinct answer patterns. After the random 80/20 split, **281 of the 301 test people (93.4%) have an identical answer pattern in the training set**. Only 20 test people present a pattern the model has not seen.

Table 9.1. Hybrid Stack test accuracy, split by whether the person's answer pattern also occurs in the training set.

Test people	Number	Share of test set	Hybrid accuracy
With a twin in the training set	281	93.4%	100.00%
Without a twin	20	6.6%	80.00%
All	**301**	**100%**	**98.67%**

The hybrid is right on every one of the 281 people with a twin and on 16 of the 20 without one. All 4 of its test errors fall in the small group of unseen patterns. A group of 20 people is too small for a precise estimate, but the direction is unmistakable: the overall test accuracy is an average dominated by cases the model has effectively memorised.

9.2 Grouped cross-validation

Grouped five-fold cross-validation gives a larger and more reliable sample of the same situation. Every answer pattern is confined to one fold, so each model is always tested on patterns absent from its training data. Table 9.2 and Figure 9.1 compare the result with standard cross-validation.

Table 9.2. Standard ten-fold cross-validation against grouped five-fold cross-validation (no answer pattern shared between training and test folds).

Model	Standard CV	Grouped CV	Drop (points)	Grouped balanced acc.
Random Forest	97.92%	86.03%	11.9	83.76%
CatBoost	98.25%	85.16%	13.1	82.78%
Hybrid Stack (proposed)	**98.00%**	**84.70%**	**13.3**	**82.60%**
XGBoost	97.83%	83.83%	14.0	82.07%
LightGBM	98.08%	83.70%	14.4	81.84%
MLP Neural Net	97.59%	80.17%	17.4	78.60%
Logistic Regression	82.70%	79.84%	2.9	77.01%

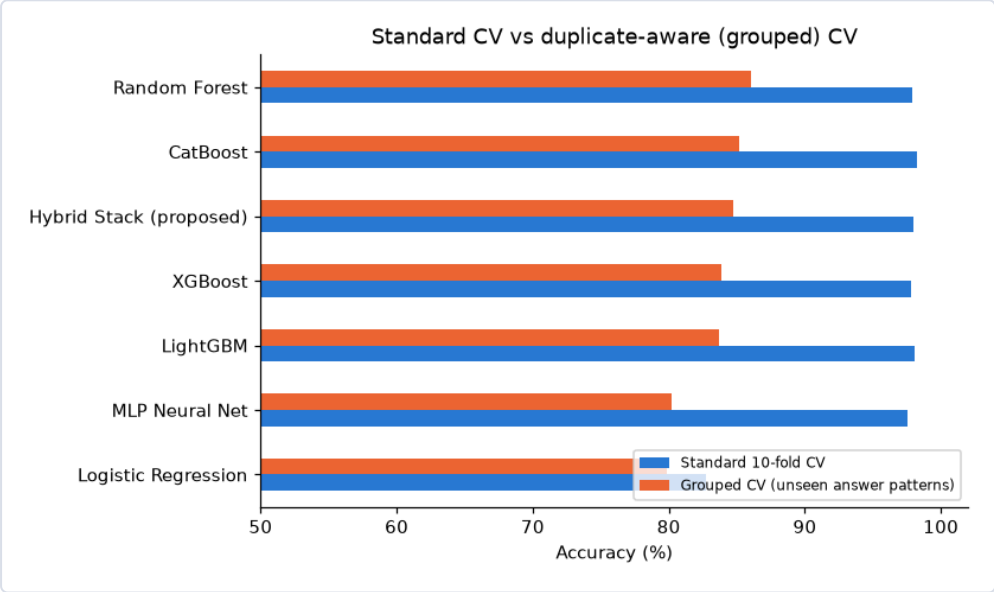


Figure 9.1. Standard cross-validation (blue) against duplicate-aware grouped cross-validation (orange).

Three results follow.

- **All six non-linear models lose 12 to 17 points.** They fall from about 98% to between 80.2% and 86.0%. The Hybrid Stack reaches 84.70%.
- **Logistic regression barely moves** (82.7% to 79.8%). A linear model cannot memorise individual patterns, so it gains nothing from seeing them twice. The gap between it and the flexible models shrinks from about 15 points to about 5.
- **The ranking changes.** Random Forest is best on unseen patterns (86.03%), and the order among the top models differs from the standard ranking. No model that looks best under the random split can be assumed best in deployment.

The balanced accuracies are 1.6 to 2.8 points below the plain accuracies, so on unseen patterns the models also become less even-handed between the two classes.

9.3 How to read a 100% score

The two numbers answer different questions, and both are correct answers to their own question.

Table 9.3. What each accuracy figure measures.

Figure	Protocol	Question it answers	Use it for
98.7–100% (test), about 98% (CV)	Random split of all rows	How often is the model right on a new row drawn from this file, most of which repeat rows it has seen?	Comparison with earlier papers, which used this protocol
80.2–86.0%	Grouped split by answer pattern	How often is the model right for a woman whose combination of answers it has never seen?	Any statement about real-world screening performance

READING CATBOOST'S 100%

Because no answer pattern in this dataset occurs with both target values, a model flexible enough to memorise the training patterns is guaranteed to be right on every test person with a twin. CatBoost does this completely and also happens to be right on the 20 people without a twin in this particular split. Its grouped accuracy of 85.16% shows what remains without the twins. A score of 100% here signals a property of the data, not a perfect classifier, and reporting it without the grouped figure would mislead.

ANSWER TO RQ5

When no answer pattern is shared between training and test data, the accuracy of the proposed model falls from 98.0% to 84.7%. Roughly 13 percentage points of the headline figure come from recognising repeated patterns. The remaining 85% is still well above the 65.2% majority-class floor and above the linear baseline, so the model has learned real structure, only less than the headline suggests.

CHAPTER 10

Comparison with Published Work

The proposed model next to the published results on the same dataset, in accuracy and in everything a paper should report besides accuracy.

10.1 Accuracy under the common protocol

Figure 10.1 and Table 10.1 place this study among the earlier ones. All figures use a random split of the full dataset, so they are comparable with each other and all share the limitation described in Chapter 9.

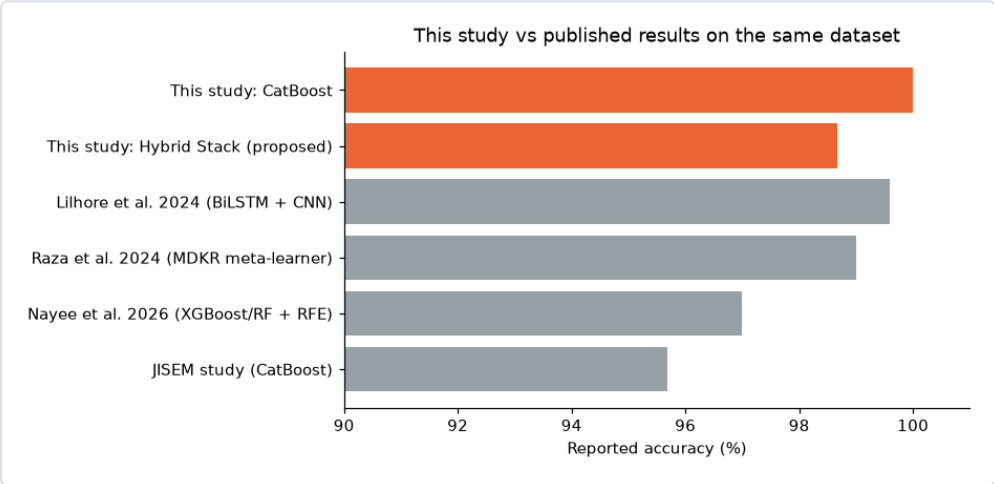


Figure 10.1. Reported accuracy of earlier studies (grey) and of this study (orange) on the same dataset.

Table 10.1. Accuracy on the dataset under a random split: earlier studies and this study.

Study	Accuracy
JISEM study (CatBoost)	95.68%
Nayee et al. 2026 (XGBoost/RF + RFE)	96.99%
This study: Hybrid Stack (proposed)	**98.67%**
Raza et al. 2024 (MDKR meta-learner)	99.00%
Lilhore et al. 2024 (BiLSTM + CNN)	99.60%
This study: CatBoost	**100.00%**

Bars start at 90%. Accuracies of earlier studies are taken from abstracts and search summaries and must be confirmed in the full texts.

The Hybrid Stack's 98.67% lies above two of the four earlier results and below the other two. Its confidence interval, 97.34–99.67%, reaches from just above the two lower published values to just above the highest. No published study reports an interval, so it is not possible to say whether any of the differences in the table are larger than chance; with test sets of about 300 people, most are probably not. This study's CatBoost result is the highest number in the table, and Chapter 9 explains why it should not be advertised as such.

10.2 What surrounds the number

The more informative comparison is of what each study reports around its accuracy (Table 10.2).

Table 10.2. What each study reports besides accuracy. Entries for earlier studies reflect their abstracts.

Study	Statistical tests / CI	Per-person XAI	XAI methods compared	Uncertainty	Counter-factual	Duplicate audit
JISEM study [5]	–	–	–	–	–	–
Nayee et al. [3]	–	–	–	–	–	–
Raza et al. [2]	–	–	–	–	–	–
Lilhore et al. [1]	–	–	–	–	–	–
Shivaprasad et al. [4] (different target)	–	Yes	shown only	–	–	–
This study	Yes	Yes	Yes	Yes	Yes	Yes

The comparison should be read with care in both directions. A dash means that the abstract does not mention the item, not that the full paper certainly lacks it. And the last column cuts both ways: the duplicate audit is this study's main methodological contribution, and it is also the result that lowers the realistic value of this study's own headline accuracy.

CHAPTER 11

Discussion

What the results mean together, which problems the project solved, and how the findings should be presented to others.

11.1 Answers to the research questions

Table 11.1. Summary of answers to the research questions.

	Answer	Key evidence
RQ1	CatBoost scores highest, but the six non-linear models are statistically indistinguishable. The Hybrid Stack matches the best single models and has better-calibrated probabilities than most.	Test 98.67% (CI 97.34–99.67%); Holm p = 1.0 against five models
RQ2	*Feeling of guilt* dominates. SHAP and LIME agree globally but differ for individuals.	Global Spearman 0.917; per-person 0.534
RQ3	SHAP is more faithful and additive; stability is similar; LIME is faster.	Top-1 faithfulness 0.301 vs 0.237
RQ4	Probabilities are calibrated, conformal sets meet their promise, a one-answer counterfactual exists, and no age gap is detectable.	Brier 0.006; coverage 97.0% at a 95% target
RQ5	About 13 points of accuracy come from repeated answer patterns.	Grouped CV 84.7% vs standard 98.0%

Taken together, the results describe a dataset on which any sufficiently flexible model reaches about 98% under a random split, because most test rows are copies of training rows, and about 85% on new patterns. Within that setting, the choice among modern models matters little for accuracy. What distinguishes a useful model is the quality of its probabilities, its explanations and its ability to abstain, and on those the Hybrid Stack performs well.

The hybrid did not beat its best base model. This is a common and honest outcome of stacking: when the base models are already near the ceiling and make almost the same errors, there is little for a meta-learner to add. What stacking did deliver is robustness. The hybrid is never the worst of the strong models on any metric, and its Brier score is better than that of three of its four base models.

11.2 What this project solved

Table 11.2 lists the problems the project set out to solve, what was done and how far each is solved.

Table 11.2. Problems addressed, the solution implemented and its status.

Problem	What was done	Status
Models compared on single, untested numbers	Ten-fold CV, four statistical procedures, bootstrap CI	Solved: differences among six models shown to be non-significant
Black-box predictions	Exact SHAP and LIME for 100 people; permutation importance for all models	Solved for this model
No basis for choosing an XAI method	Four-criterion comparison of SHAP and LIME	Solved for this model and data; SHAP preferred
Overconfident predictions	Calibration analysis and conformal prediction at three levels	Solved: coverage promise met
Explanations that are not actionable	Exhaustive counterfactual search	Demonstrated on one person; not yet systematic
Unexamined subgroups	Accuracy per age band	Partly solved: only age is available
Hidden leakage from repeated rows	Twin analysis and grouped cross-validation	Solved as a measurement; the dataset itself cannot be repaired
Irreproducible pipelines	One notebook, fixed seed, pinned versions, results regenerated on every run	Solved

11.3 Practical implications

- **For researchers using this dataset.** Report grouped validation next to the random split. A new architecture that moves random-split accuracy from 98% to 99% has shown very little; one that moves grouped accuracy from 85% to 90% would have shown something real.
- **For anyone building a screening aid.** The six non-linear models make no false positives and a few false negatives at the default threshold of 0.5. In screening, a missed case costs more than a false alarm, so the threshold should be chosen deliberately, on validation data, to reach a target recall.
- **For users of explanations.** Two popular methods name a different main reason for a substantial minority of people. An explanation shown to a clinician should come from the more faithful method and should not be presented as a fact about the patient.
- **For interpreting feature importance.** The dominant question behaves in this dataset in a way that contradicts clinical expectation. A model can be accurate and well explained and still encode a relationship that would not hold elsewhere.

11.4 How the results should be reported

When this work is written up for a journal or presented, the following order keeps the claims in proportion to the evidence.

1. Lead with the cross-validation result as mean and standard deviation: Hybrid Stack 98.00% ± 1.26; CatBoost 98.25% ± 1.44. Ten splits are more reliable than one.

2. Give the single test-set result with its confidence interval, including CatBoost's 100%, and state in the same paragraph that 93.4% of test people have a twin in the training set.
3. Give the grouped cross-validation result (80.2–86.0%) as the estimate for unseen answer patterns.
4. State that the proposed model is not significantly more accurate than the best single models, and present explanation quality, calibration, uncertainty and the duplicate audit as the contributions.

A SENTENCE THAT CAN BE USED AS WRITTEN

"Under the random-split protocol of earlier work, the proposed stacking ensemble achieved 98.67% test accuracy (95% CI 97.34–99.67%) and 98.00% in ten-fold cross-validation, statistically indistinguishable from CatBoost and random forest. Because 78% of the records repeat an earlier answer pattern, we additionally report grouped cross-validation, under which accuracy was 84.7%."

CHAPTER 12

Limitations and Threats to Validity

What this study cannot show, stated as plainly as what it can.

12.1 Data limitations

- **The label is not a diagnosis.** The target is one self-reported questionnaire item. The model predicts how a woman answers one question from how she answers nine others. That is related to, but not the same as, detecting an anxiety disorder.
- **Provenance is undocumented.** The dataset's listing gives little detail on how, where and from whom the data were collected. All timestamps fall within 26.4 hours, and 78% of rows are repeats. Whether the repeats are genuine identical answers, duplicated entries or generated records cannot be determined from the file.
- **The effective sample is small.** There are 326 distinct answer patterns, not 1,503 independent observations. Grouped validation estimates performance from these.
- **A counter-intuitive main predictor.** The strongest question relates to the target in the opposite direction from clinical expectation, which raises doubts about how the item was worded, coded or answered.
- **Few features and one demographic.** There is no information on parity, time since birth, history of mental illness, social support, income or region, so neither richer prediction nor a serious fairness analysis is possible.
- **No external data.** All results come from one file. Nothing is known about how the model would perform on women from another hospital or country.

12.2 Method limitations

- **One train/test split.** The final test uses a single split of 301 people. The confidence interval reflects sampling of test people but not the variation that a different split would produce.
- **Tuning and model selection used random splits.** Optuna and the ten-fold comparison ran on random folds, which contain twins. The selected hyper-parameters may therefore favour memorisation. Grouped validation was applied only at the end, as a check, with the same settings.
- **Small tuning budget.** Only two models were tuned, with 25 trials each. The others use fixed settings, so the comparison is between reasonable configurations, not between each model's best possible form.
- **Explanations cover 100 people and one model.** The SHAP-versus-LIME findings apply to the Hybrid Stack on this dataset. Stability was measured on 20 people.

- **Faithfulness is measured in one way.** The deletion test replaces answers with those of background people, which is the same idea SHAP itself is built on and may favour it.
- **Conformal calibration uses out-of-fold scores from random folds.** The coverage guarantee assumes that calibration and test data are exchangeable, which holds for the random split but would not hold for women with unseen patterns.
- **One counterfactual example.** It illustrates the method and is not a systematic evaluation.
- **Literature numbers are second-hand.** They come from abstracts and summaries and several details are unconfirmed.

12.3 Threats to validity

Table 12.1. Threats to validity and what was done about each.

Type	Threat	Mitigation in this study
Internal	Leakage through preprocessing	Encoding inside pipelines; tuning on training data only; test set used once
Internal	Leakage through repeated rows	Measured with twin analysis and grouped CV; reported next to standard results
Statistical	Chance differences read as real	Friedman, Wilcoxon–Holm, McNemar, bootstrap CI
Statistical	Fold accuracies are not independent	Non-parametric tests; conclusions drawn only from large effects
Construct	Self-reported item used as outcome	Stated throughout; no clinical claim made
Construct	Importance mistaken for causation	Explanations described as properties of the model; counter-intuitive direction reported
External	Results do not transfer to other populations	Not mitigated; external validation is future work
Reproducibility	Results depend on software versions	Versions pinned; seed fixed; book generated from result files

CHAPTER 13

Future Work

What should come next, ordered by how soon it can be done.

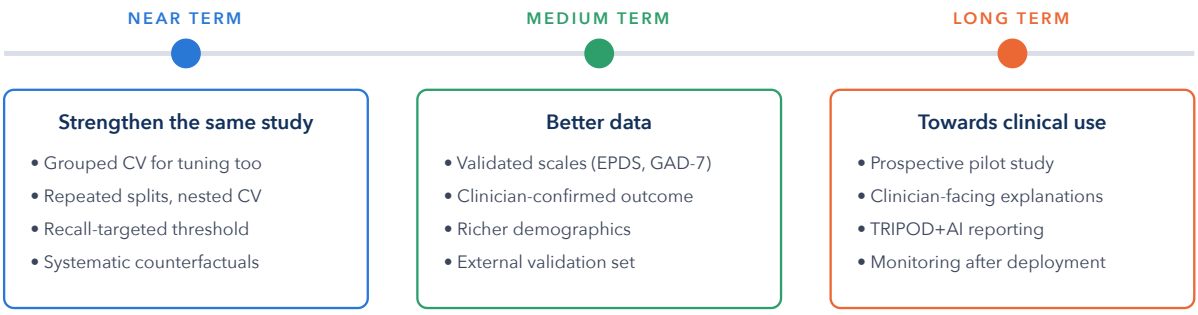


Figure 13.1. Roadmap for future work.

13.1 Near term: strengthen the same study

- **Use grouped validation from the start.** Tune hyper-parameters and select models with grouped folds, so that the pipeline is optimised for unseen patterns and not for memorisation. Simpler, more regularised models may then win.
- **Repeat and nest.** Replace the single test split with repeated grouped cross-validation, and nest the tuning inside it, to obtain confidence intervals that include split-to-split variation.
- **Choose the threshold for screening.** Select the decision threshold to reach a stated recall, for example 95%, and report the specificity that results.
- **Evaluate counterfactuals systematically.** Compute them for every flagged person and report how many answers must change, which answers change most often and how plausible the changes are.
- **Broaden the XAI comparison.** Add further methods such as Anchors and model-specific TreeSHAP, a second faithfulness measure that does not rely on background replacement, and explanations for the other strong models.
- **Deduplicated analysis.** Train and test on the 326 unique patterns alone, with pattern frequency as a weight, as a third view next to the random and grouped protocols.
- **Conformal prediction under shift.** Calibrate with grouped folds and examine whether coverage holds for unseen patterns.

13.2 Medium term: better data

- **A dataset with documented provenance.** Collect or obtain data with a clear protocol, informed consent, one record per woman and a known collection period.

- **Validated instruments.** Use established scales such as the Edinburgh Postnatal Depression Scale and the GAD-7, whose scores have known clinical meaning, instead of single ad-hoc items.
- **A clinical reference standard.** Take the outcome from a clinician's assessment, so that the model predicts a diagnosis and not another questionnaire answer.
- **Richer predictors.** Add time since birth, parity, obstetric complications, psychiatric history, social support and socio-economic variables. These also make a real fairness analysis possible.
- **External validation.** Test the frozen model on data from a different site before making any claim about generalisation.

13.3 Long term: towards clinical use

- **Prospective evaluation.** Run the model alongside routine screening in a pilot and compare its referrals with clinicians' decisions, without letting it influence care.
- **Explanations designed with clinicians.** Study which form of explanation (contribution bars, counterfactual sentences, uncertainty flags) clinicians understand and find useful.
- **Reporting and governance.** Follow TRIPOD+AI [28] for reporting and trustworthy-AI guidance such as FUTURE-AI [29] for design; obtain ethical approval; plan for monitoring of performance and drift after deployment.
- **Local-language deployment.** A screening aid for use in Bangladesh or similar settings needs questionnaires validated in the local language and culture, and an interface usable by community health workers.

CHAPTER 14

Conclusion

The study in four paragraphs.

This project set out to do more than add another high accuracy to a dataset that already has several. It built seven models under one protocol, proposed a heterogeneous stacking ensemble, and then examined that ensemble from every side a screening tool should be examined from.

Under the random-split protocol of earlier work, the Hybrid Stack reached 98.67% test accuracy (95% CI 97.34–99.67%) and 98.00% in ten-fold cross-validation, inside the published range of 95.68% to 99.6%. Statistical tests showed that it is clearly better than a linear baseline and not distinguishable from CatBoost, random forest, LightGBM, XGBoost or a neural network. Its probabilities are well calibrated, its conformal prediction sets meet their coverage promise, and both SHAP and LIME identify the same dominant question, with SHAP giving the more faithful per-person explanations.

The most consequential finding concerns the data. Only 326 of the 1,503 responses are distinct, 93.4% of test people have an exact twin in the training set, and when no pattern is shared between training and test data, accuracy falls to 80.2–86.0%. The 98–100% figures reported on this dataset, here and elsewhere, mostly measure recognition of answers already seen. About 85% is the realistic estimate for a new answer pattern.

The lasting contribution is therefore methodological. Accuracy on this benchmark should be reported with a duplicate-aware validation, a confidence interval and a statistical test; explanations should be measured and not only shown; and a screening model should be able to say when it does not know. With better data, the same framework can be carried from a public benchmark towards a tool that clinicians could trust.

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APPENDIX A

Reproducing the Study

Everything needed to regenerate every number, figure and table in this book.

Environment

Python 3.11 with the pinned versions of Table 5.5. From the project folder:

```
python3.11 -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
cd notebooks && jupyter lab postpartum_anxiety_xai_study.ipynb   # Run All
```

Select the project's `.venv` as the notebook kernel. Running on a different Python or library version changes a few numbers slightly (for example the XGBoost test accuracy), because tree-building and random-number details differ between versions. The run takes a few minutes and rewrites every file in `results/`.

Rebuilding this book

```
pip install jinja2 pandas pymupdf
python book/build_book.py
```

The build script reads the result tables, the raw data file and the printed output of the executed notebook, fills them into the book template and prints it to PDF with Google Chrome. No number in the book is typed by hand.

Project layout

Table A.1. Folders of the project.

Path	Contents
`data/raw/`	The questionnaire file, never modified
`data/DATA_CARD.md`	Source, licence, known issues
`notebooks/`	The study notebook: 57 tasks in 10 parts
`results/`	Figures (PNG) and tables (CSV) written by the notebook
`research/`	Literature review and research gaps
`book/`	This book, its template and its build script
`requirements.txt`	Pinned library versions

APPENDIX B

Notebook Task Map

Every task of the study notebook, in order, so that any result in this book can be traced to the cell that produced it.

Part	Task	What it does
Part A · Setup	1	Import the libraries
	2	Settings
	3	Load the dataset
Part B · Understanding the data	4	Column types and possible answers
	5	Missing answers
	6	Repeated answer patterns
	7	Clean the data
	8	Figure : class balance
	9	Figure : anxious rate for each answer to one question (guilt)
	10	Figure : strength of association with the target (Cramér's V)
Part C · Preparing the data	11	Turn answers into numbers
	12	Train / test split (80% / 20%, stratified)
	13	A small helper : one-hot encoding inside a pipeline
Part D · The seven competing models	14	Model 1 : Logistic Regression
	15	Model 2 : Random Forest
	16	Model 3 : XGBoost, tuned with Optuna
	17	Model 4 : LightGBM, tuned with Optuna
	18	Model 5 : CatBoost
	19	Model 6 : MLP neural network
	20	Model 7 : the proposed Hybrid Stack
	21	List of competitors
	22	Figure : how Optuna improved over its trials
Part E · Cross-validation competition	23	10-fold stratified cross-validation on the training set
	24	Cross-validation table (mean ± standard deviation)

Part	Task	What it does
	25	Figure : cross-validation accuracy of every model
Part F · Final test on the held-out 20%	26	Train every model on the full training set
	27	A function that computes all metrics
	28	Test-set results table
	29	Figure : test accuracy of every model
	30	Overfitting check : train accuracy vs test accuracy
	31	Figure : ROC curves
	32	Figure : confusion matrix of the Hybrid Stack (and a table for every model)
	33	Figure : calibration of the Hybrid Stack
	34	Statistical tests on the 10 cross-validation folds
	35	McNemar test and bootstrap confidence interval on the test set
Part G · Explainable AI (SHAP and LIME)	36	Prepare the people to explain and the prediction function
	37	Compute SHAP values
	38	Figure : SHAP global importance
	39	Figure : SHAP effect of each answer to the most important question
	40	Figure : SHAP explanation of one person
	41	Compute LIME explanations
	42	Figure : LIME global importance
	43	Figure : LIME explanation of the same person
	44	SHAP vs LIME (1) : do they rank the questions the same way?
	45	SHAP vs LIME (2) : agreement
	46	SHAP vs LIME (3) : faithfulness (deletion test)
	47	SHAP vs LIME (4) : stability
	48	SHAP vs LIME : summary table
	49	Table : do the seven models agree on what is important? (permutation importance)
Part H · Trustworthy-AI extras	50	Conformal prediction : an honest "I am not sure"
	51	Figure : conformal coverage
	52	Counterfactual explanation : "what would have to change?"
	53	Fairness check : accuracy in every age group

Part	Task	What it does
Part I · Robustness check: duplicate-aware validation	54	How many test people have a twin in the training set?
	55	Grouped 5-fold cross-validation (no answer pattern is shared between training and test)
Part J · Comparison with published work and final summary	56	Comparison with earlier papers on the same 1,503-row dataset
	57	One table with the headline numbers

APPENDIX C

Complete Result Tables

One table with every model under every protocol, and the headline numbers of the study.

Table C.1. Every model under every protocol (accuracy in percent).

Model	Train	10-fold CV	Test	Grouped CV	Test errors	AUC	Brier
CatBoost	100.00	98.25 ± 1.44	100.00	85.16	0	1.000	0.002
Random Forest	100.00	97.92 ± 1.37	99.34	86.03	2	1.000	0.014
MLP Neural Net	100.00	97.59 ± 1.39	98.67	80.17	4	0.997	0.015
Hybrid Stack (proposed)	**100.00**	**98.00 ± 1.26**	**98.67**	**84.70**	**4**	**1.000**	**0.006**
LightGBM	100.00	98.08 ± 1.47	97.67	83.70	7	1.000	0.022
XGBoost	100.00	97.83 ± 1.53	97.01	83.83	9	0.995	0.020
Logistic Regression	83.44	82.70 ± 2.34	81.73	79.84	55	0.895	0.122

Table C.2. Headline numbers of the study.

Quantity	Value
Responses / distinct answer patterns	1,503 / 326
Anxious / not anxious	980 / 523
Training / test people	1,202 / 301
Best model on the test set	CatBoost (100.00%)
Hybrid Stack test accuracy (95% CI)	98.67% (97.34–99.67%)
Hybrid Stack F1 / recall / specificity	98.97% / 97.96% / 100.00%
Hybrid Stack ROC-AUC / Brier / MCC	1.000 / 0.006 / 0.971
Hybrid Stack 10-fold CV accuracy	98.00% ± 1.26
Hybrid Stack grouped-CV accuracy	84.70%
Friedman test over seven models	$\chi^2 = 37.68$, $p < 0.001$
Test people with a twin in the training set	281 of 301 (93.4%)
Top question (SHAP and LIME)	Feeling of guilt
SHAP–LIME Spearman, global / per person	0.917 / 0.534
Conformal coverage at 90 / 95 / 99% target	94.02% / 97.01% / 100.00%

APPENDIX D

Glossary and Abbreviations

Terms used in this book, in plain language.

Term	Meaning
Answer pattern	One complete combination of answers to the nine feature questions
AUC / ROC-AUC	Area under the receiver operating characteristic curve; 1.0 is perfect ranking, 0.5 is chance
Bagging	Training many models on resampled data and averaging them
Balanced accuracy	Mean of recall and specificity
Base model	A model whose predictions feed the meta-learner in a stack
Boosting	Building models in sequence, each correcting the errors of the previous ones
Bootstrap	Resampling the data with replacement to estimate the uncertainty of a statistic
Brier score	Mean squared error of predicted probabilities; lower is better
Calibration	Agreement between predicted probabilities and observed frequencies
CI	Confidence interval
Conformal prediction	A method that returns a set of answers with a guaranteed coverage rate
Counterfactual	The smallest change to the inputs that would change the prediction
Coverage	Share of cases in which a prediction set contains the true answer
Cramér's V	Strength of association between two categorical variables, from 0 to 1
CV	Cross-validation: repeated training and testing on different parts of the data
Data leakage	Information from test data reaching the model during training
EPDS	Edinburgh Postnatal Depression Scale, a validated screening questionnaire
Faithfulness	How well an explanation reflects what the model actually does
Friedman test	Non-parametric test for differences among several models measured on the same folds
GAD-7	Generalised Anxiety Disorder 7-item scale
Grouped CV	Cross-validation in which all rows of a group stay in the same fold
Holm correction	Adjustment of p-values when several tests are made
Hybrid Stack	The proposed model: CatBoost, XGBoost, random forest and MLP combined by logistic regression
LIME	Local Interpretable Model-agnostic Explanations

Term	Meaning
MCC	Matthews correlation coefficient
McNemar test	Test comparing two models on the cases where they disagree
Meta-learner	The model that combines the base models' predictions in a stack
MLP	Multilayer perceptron, a feed-forward neural network
One-hot encoding	Representing each possible answer as its own 0/1 column
Out-of-fold prediction	A prediction for a person made by a model that was not trained on that person
Permutation importance	Loss of accuracy when one feature is shuffled
Recall (sensitivity)	Share of true positives that the model finds
SHAP	SHapley Additive exPlanations
Specificity	Share of true negatives that the model correctly clears
Stacking	Combining several models by training another model on their predictions
Stratified split	A split that keeps the class proportions equal in every part
Twin	A test person whose exact answer pattern also occurs in the training set
Wilcoxon signed-rank test	Non-parametric paired test for two models measured on the same folds
XAI	Explainable artificial intelligence